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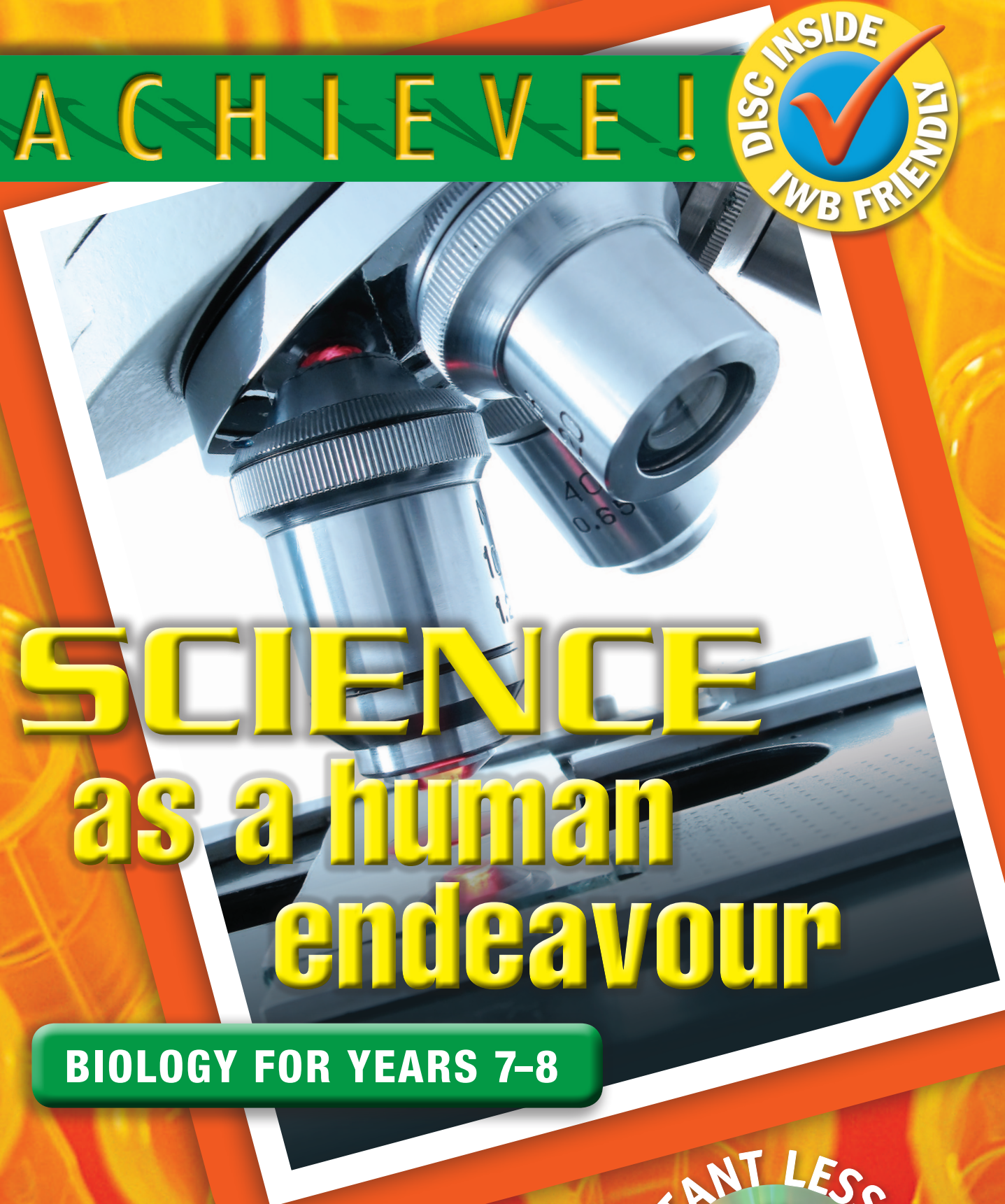

SCIENCE as a human endeavour

BIOLOGY FOR YEARS 7–8

Keith McTaggart

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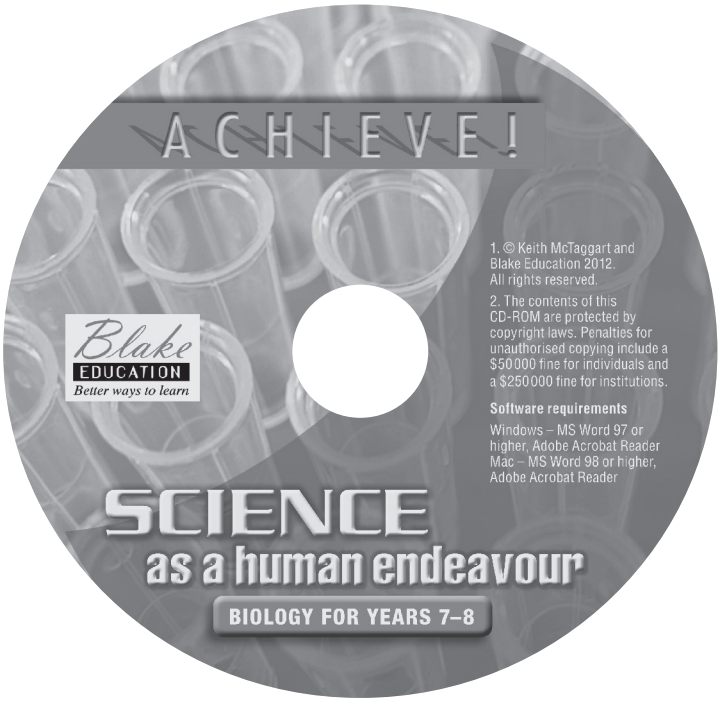

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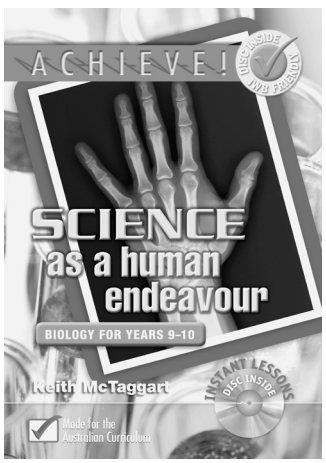

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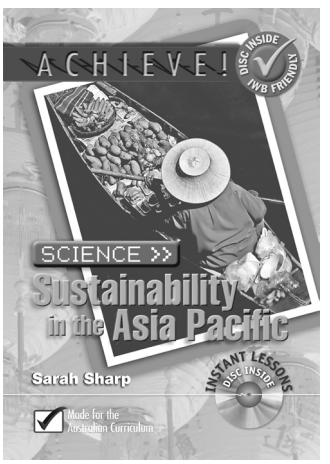

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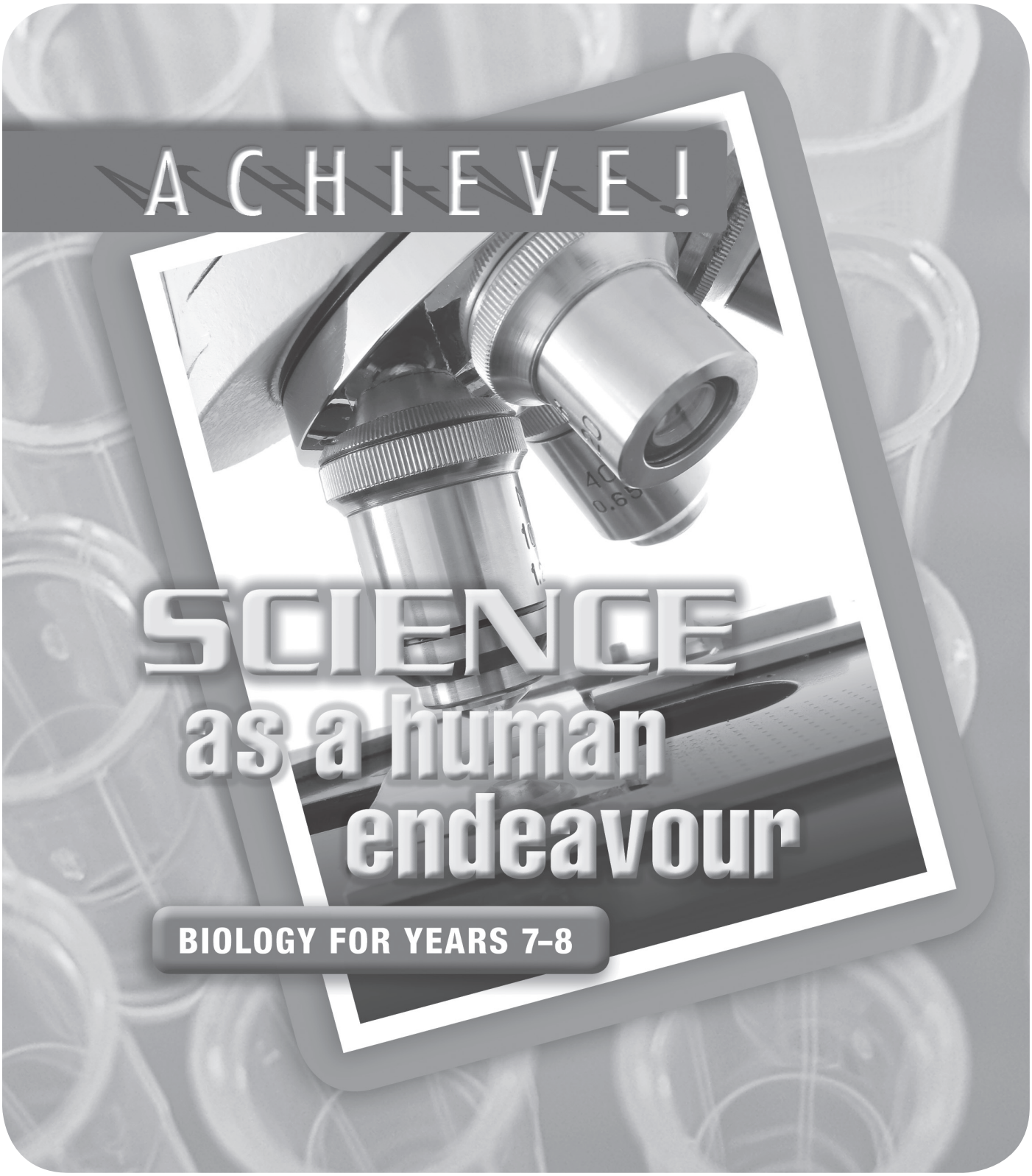

Keith McTaggart

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Project management and editing by Vanessa Lanaway
Typesetting by digiXL
Cover design by Polar Design

Keith McTaggart worked in state schools in Victoria for more than 30 years as a biology and science teacher, regional science consultant, leader of student management teams, curriculum teams and professional learning teams and as Assistant Principal. He has a science degree and diploma of education from the University of Melbourne and a diploma of freelance journalism.

Keith believes that science is an integral part of human activity in society, with scientific skills such as careful observation and principles such as evidence-based decision-making playing a large role in everyday life.

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An imprint of Blake Education Pty Ltd
ABN 50 074 266 023
108 Main Road
Clayton South VIC 3169
Ph: +61 3 9558 4433
info@blake.com.au
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Introduction

Achieve!

The *Achieve!* series of Science titles has been developed for teachers who require modified, curriculum-focused activities for secondary school students who struggle to read and comprehend regular classroom materials because of poor literacy skills. Many of the pages will also be suitable as supplementary material for regular classes.

This book will assist teachers to develop lessons for the biology sub-strand in *Science as a human endeavour* for years 7 and 8. Units of work cover a variety of topics including water management, human impacts on the environment, developments in medical and agricultural technology and the ethical questions these present, waste treatment, international cooperation in science, and developing personal responses to environmental issues. Work sheets help students build their scientific language knowledge and understanding of key concepts.

The CD-ROM contains all the pages that are featured in the book, as well as a file of illustrations that can be used for other work sheets. These documents can be easily edited to suit the individual needs of students.

This book is divided into ten units. The units can be taught as a lesson with students in groups of varying sizes. Alternatively, a single Resource or Activity sheet can be used as support material, for individual homework, assessment or revision.

The teacher's notes provide valuable guidance about using each unit, and contain the following elements.

Objectives – These are the key skills and knowledge learnt through using the work sheets.

Prior knowledge – This refers to the skills and knowledge required for students to complete the tasks. Some activities are more challenging than others, however, as a rule, students should have a reading comprehension age of six to nine years to be able to attempt the activities.

Background – This section provides additional guidance to help teachers to present a successful lesson. It may include definitions or further background notes about the subject matter.

Starter activity – Warm-up activities are provided to put the content into proper context and to gain student interest and attention.

Resource and Activity work sheets – The Resource sheets do not always contain activities and are used either to stimulate discussion or as a part of the unit's activities. The Activity work sheets may be used sequentially or as stand-alone tasks, depending on the needs of the student.

Extension ideas – Additional tasks are suggested to recap on the main skills or knowledge taught in the lesson, as well as ways to extend, assess or include technology in the lesson.

The Australian Curriculum correlations

	Year 7 / Year 8	Units 1	2	3	4	5	6	7	8	9	10
Science as a human endeavour **Nature and development of science**											
Scientific knowledge changes as new evidence becomes available, and some scientific discoveries have significantly changed people's understanding of the world	ACSHE119			✓	✓						
	ACSHE134	✓		✓	✓	✓		✓	✓	✓	
Science knowledge can develop through collaboration and connecting ideas across the disciplines of science	ACSHE223	✓	✓				✓	✓	✓	✓	✓
	ACSHE226			✓	✓			✓	✓	✓	
Use and influence of science											
Science and technology contribute to finding solutions to a range of contemporary issues; these solutions may impact on other areas of society and involve ethical considerations	ACSHE120	✓	✓	✓	✓		✓	✓	✓	✓	✓
	ACSHE135	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
Science understanding influences the development of practices in areas of human activity such as industry, agriculture and marine and terrestrial resource management	ACSHE121	✓	✓				✓	✓	✓	✓	✓
	ACSHE136	✓	✓							✓	✓
People use understanding and skills from across the disciplines of science in their occupations	ACSHE224	✓							✓		
	ACSHE227		✓	✓	✓	✓		✓	✓	✓	✓

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Teacher's notes – Water management

Achieve!

Objectives

- To learn that water management is the responsibility of individuals, businesses and governments
- To understand that individual action is needed in order for national management to succeed
- To learn how to interpret and construct tables of information
- To learn how to interpret diagrams and illustrations
- To be able to interpret flowcharts

Prior knowledge

Students should know that Australia is a dry continent and water management is important. Students should also understand that humans need clean water every day, and that this need is second only to the need for oxygen in order to survive.

Background

Water management is a national problem. People at every level of society are responsible, from the federal government to individuals. Every citizen needs to use water responsibly. Individual water conservation and community-wide water reclamation are both important.

Starter activity

As a class, brainstorm uses of water, types of water and places water finishes up after use. List these on the whiteboard or on butcher's paper. Hold a class discussion about the many ways that water is used and whether students consider these uses justified or unnecessary. Encourage students to consider different points of view. For example, inner city children may not feel that irrigation is necessary and rural students may not understand opposition to irrigation.

Activity sheets

The Activity sheet 'Different types of water' introduces different types of water and possible uses of each type. Students need to decide which types are safe to use and then complete a table demonstrating their understanding.

The 'Indigenous use of water' Activity sheet outlines how the CSIRO consults with Indigenous people about local water courses in various parts of Australia. Creeks or rivers have spiritual as well as practical uses. The location of various activities around a campsite is important, and students are asked to decide the best position for various activities – upstream of the camp, near the camp, downstream or well away from the creek bank. A vowel-replacing activity completes the lesson.

'Irrigation – Benefits and challenges' introduces the idea that there are both benefits and problems associated with irrigation. Students match problems with possible solutions.

In the Activity sheet 'Water use and abuse', students assess the water use of a family and identify water-saving solutions.

The 'Desalination' Activity sheet introduces desalination as a way of increasing fresh water supplies. A flowchart demonstrates how desalination works, and students complete a cloze activity interpreting the information.

Extension ideas

- A class trip to a local creek may help students to visualise the campsite when completing the 'Indigenous use of water' Activity sheet.
- To explore the topic of how Indigenous people use water, go to the CSIRO water website – see www.csiro.au/resources/Indigenous-Water-Values-Katherine for example. The work sheet on page 12 also explores the topic further.

Suggested web addresses were functional at the time of publication but teachers should check them for inappropriate content and to ensure that they are still functional before use with students.

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Different types of water

In many parts of Australia, water is often in short supply. There are different types, or grades, of water, and not all of them are safe to drink. **Drinking water** can be used for anything but should not be wasted. **Storm water** is the water that runs off after a heavy rainfall. It is safe to drink if carefully stored. **Grey water** is not safe for drinking and cannot be stored for long, but can be used to water gardens. **Black water** is dangerous and smelly until after it has been **treated**.

☞ **1** Fill in the table below by writing 'yes' or 'no' in the 'Can it be used?' column. Decide what each type of water can be used for and copy the correct use into the last column.

watering gardens or pastures

any use within water restrictions

drinking, washing or watering garden if safely stored

Type of water	Comes from	Can it be used?	Can be used for
Drinking water	taps, bottles or tanks		
Storm water	runs off roof after rainfall, into closed tank		
Grey water	washing machine, showers		
Black water	toilets or kitchen sink		No use before treatment – must go straight to sewerage system, septic tank or composting toilet

2 Choose the correct option from each pair in brackets.

Black water is [dangerous/safe] because it contains millions of [germs/vitamins].

Kitchen waste water has food particles, which could grow [bacteria/garden plants], so it goes to the sewage.

Grey water is safe for a while but grows germs if [used/stored], so needs to be used for watering within a day.

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Indigenous use of water

Indigenous Australians follow simple, scientific rules when setting up camp. The rules tell them how to use the water in the creek for everything they need without getting sick.

☞ **1** Imagine you are camping in a creek valley with your family. Plan your campsite. Decide where each of these actions would take place and list them in the table below.

- collect drinking water
- collect cooking water
- empty out cooking water
- wash bodies (bathe)
- let children swim
- let pets swim
- wash clothes
- wash dishes
- catch fish
- use toilet pit

Upstream	Near camp	Downstream	Away from creek

2 Why should washing and swimming take place downstream from where drinking water is collected?

_____

_____

3 Fill in the missing vowels (a, e, i, o, u) in the passage below.

CSIRO sc__ __nt__sts such as D__ct__r Sue Jackson and Brad Moggridge w__rk with Indigenous people to l__ __rn about m__n__g__ng rivers. The steady fl__w of a river brings fo__d and wat__r and a sense of w__llb__ __ng to Indigenous A__str__l__ __ns. The flow of the water has pr__ctical and spiritu__l importance, and __nd__g__n__ __s Australians h__ve learned how to manage their imp__ct on str__ams.

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Irrigation – Benefits and challenges

Australia is a dry continent. However, some parts of Australia, such as far north Queensland, sometimes receive very high rainfall. When this happens, the water flowing along rivers is collected in **reservoirs** and directed to dry areas. This water is used for crops in a process called **irrigation**. Irrigation means that instead of only growing grass and tough shrubs, farmers have enough water to grow citrus and other fruit trees, grass for dairy cattle, grapes, rice and vegetables. Although irrigation helps farmers to grow crops, it also creates some problems.

☞ **1** Match each irrigation problem listed below with its possible solution.

Problem	Possible solution
The open channels and sprinkler systems used for irrigation mean water is wasted due to **evaporation**.	Governments make rules about how much water can be taken from a river.
Some land is too dry for crops to grow, even with irrigation, so the water is wasted	Farmers measure soil moisture to reduce overwatering.
More water is used on crops than is needed.	Mulch and drip systems replace water sprinkling; pipes are used instead of open channels.
Too little water makes it downstream for other uses or for the health of a river.	Measure salinity and plant salt-removing vegetation near irrigated land.
Salt is left behind when water evaporates, causing high **salinity**.	Water authorities only allow irrigation to be used on some crop types.
Too much water is used on crops that don't give a good return.	The federal government buys back extremely dry land from farmers.

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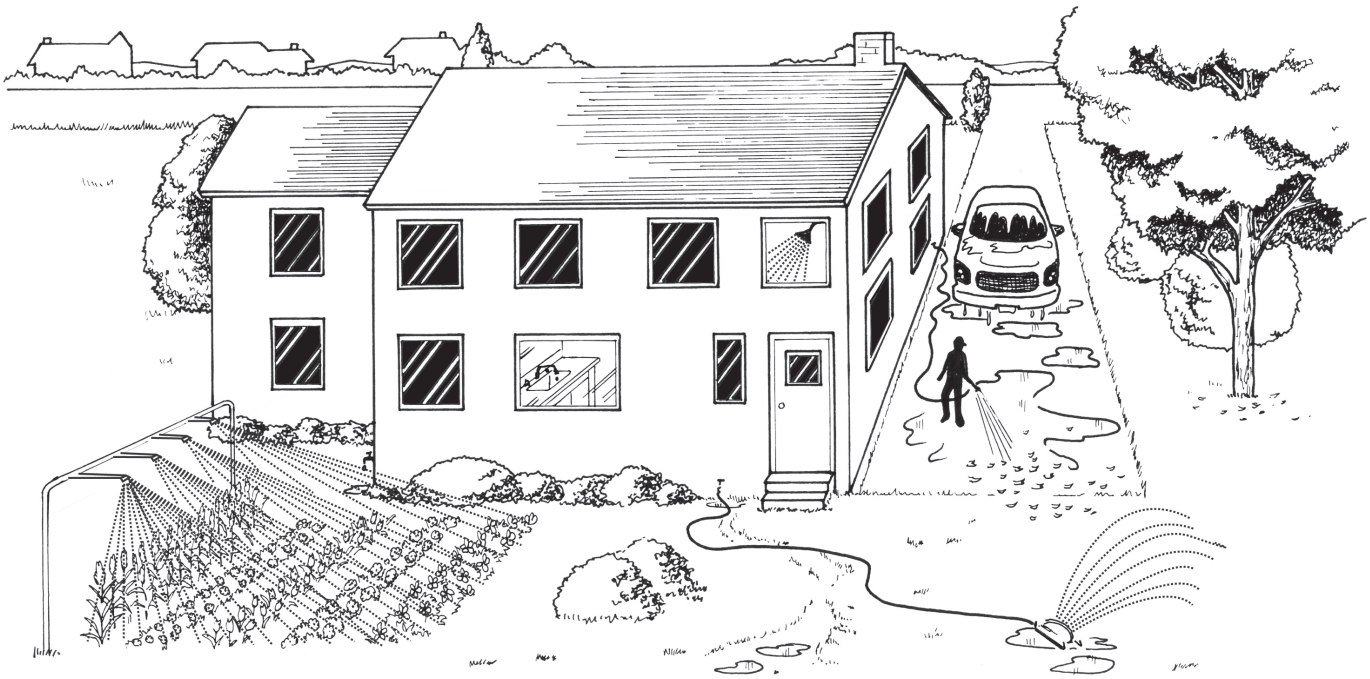
Water use and abuse

The Wakefield family needs advice about how they can stop wasting water.



☞ **1** The drawing above shows some ways the Wakefields use water. Can you find six water-wasting acts that are taking place?

a) _____ **d)** _____

b) _____ **e)** _____

c) _____ **f)** _____

2 To reduce their water use, the Wakefields need to make some changes. Join these sentence parts together to create six things the Wakefields could do to save water.

Fit new washers	a mulched garden.
Get a free water-saving shower head	with a broom.
Replace the lawn with	tolerant plants.
Plant drought-	toilet cistern.
Buy a dual-flush	to taps.
Sweep the driveway	from the local council.

3 Number the actions listed above from the cheapest to the most expensive.

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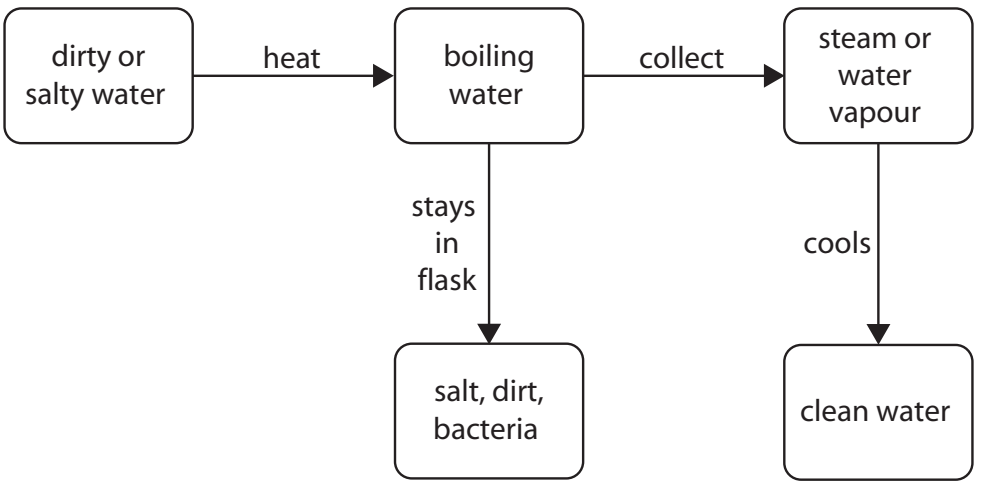
Desalination

Humans need **fresh water** to survive – we cannot drink dirty or salty water. After a disaster or during a drought, **desalinators** are used to supply fresh, clean water. The desalinator removes the dirt or salt from the water and **purifies** the water so it is clean enough to drink. The Australian Defence Force has large transportable desalinators, which can be used for disaster relief or wherever troops are stationed.

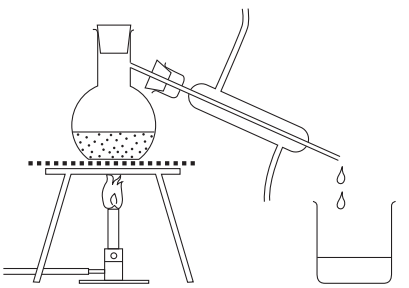
A Liebig's Condenser desalinates water on a smaller scale in a laboratory. The flowchart below shows how it works.



☞ **1** Use the word bank to complete the sentences explaining how Liebig's Condenser works.

Word bank					
flask	vapour	water	liquid	glass	bacteria

Dirty or salty water is boiled in a _____. Water _____ rises into a _____ tube. Salt, dirt and _____ stay in the flask. The _____ vapour cools and turns into _____. Droplets of clean water collect in the beaker.



Liebig's Condenser

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Teacher's notes – Human impacts

Achieve!

Objectives

- To recognise that humans have an impact on the environment in many different ways
- To recognise that humans have already inflicted environmental damage, which needs to be cleaned up
- To see that cleaning up the Earth needs to be undertaken by governments and individuals, and everyone in between
- To understand that the CSIRO is seeking help with the environment from Indigenous Australians
- To learn that scientists are improving crops in order to feed the world's population
- To learn how to interpret tables of information, flowcharts and pie charts

Prior knowledge

Students should understand that humans have an impact on the environment and that we need to improve the way we treat the Earth. Students will also need some knowledge of flowcharts before attempting to complete one in 'Waste recycling'.

Background

When asked about ways we can damage the environment, students may offer examples such as littering or shooting animals. There are many other ways we affect our surroundings, such as growing crops, clearing land for houses and producing carbon dioxide. There are huge efforts being undertaken through international scientific cooperation to repair damage caused by early researchers and explorers. In Antarctica, for example, the Protocol on Environmental Protection to the Antarctic Treaty (1991) has prompted major changes to the way the land is used.

Starter activity

Ask students to discuss individual ways of making mess or rubbish. Can anyone tell me where there is a mess? Why is it a mess? Who made the mess and who should clean it up? Expand the discussion to look at large-scale messes that damage the environment, such as oil spills or deforestation. What changes can an individual person make to reduce their own mess? How does this compare with what needs to happen in places like Antarctica or the Amazon valley?

Activity sheets

The Activity sheet 'Wetland rescue' explains how Indigenous people are using their traditional knowledge to help solve environmental problems. Students consider the benefits of steps being taken to solve problems in Kakadu National Park.

The 'Antarctica' Activity sheet discusses the international cooperation that is taking place in order to repair environmental damage in Antarctica. Students are asked to rank tasks in order of importance.

In 'Cleaning up Australia', students are asked to interpret a pie chart of rubbish collected on a Clean Up Australia Day, considering different kinds/sources of rubbish and how this can be reduced.

The Activity sheet 'Crop technology' examines the impact of our population and how we produce food. Students complete a cloze activity examining how Australian scientists are improving crops and reducing their impact on the environment.

'Waste recycling' requires students to complete a flowchart demonstrating how waste can be recycled. This activity makes students aware of the need for recycling.

Extension ideas

Complete a 'What can we do?' activity with the class. The following websites offer a host of research opportunities – there are many educational activities and webcams on the Antarctica site.

- www.csiro.au/science/KakaduWetlandBurning
- www.antarctica.gov.au
- www.cleanup.org.au/au/RubbishReport/rubbish-report

Suggested web addresses were functional at the time of publication but teachers should check them for inappropriate content and to ensure that they are still functional before use with students.

Selected answers

Wetland rescue Action 1: 1; Action 2: 2, 3; Action 3: none; Action 4: 4; Action 5: 5; Action 6: 5, 6; Action 7: 6; Action 8: 7, 8.

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Wetland rescue

Until the 1980s, Feral Water Buffalo were allowed to graze in the Northern Territory's Kakadu National Park. They damaged native wildlife and their hard hooves compressed the ground in many places. After the Feral Water Buffalo had been removed from the area, a native grass called Mudja grew wildly and choked off wetlands. Local people had previously burned the Mudja grass so other plants could grow. Senior traditional owner Violet Lawson taught CSIRO Indigenous research officers Sandra McGregor – her daughter – and Peter Christophersen the burning method.

☞ **1** Decide which benefits apply to each action in the table below, and write the matching numbers in the right-hand column. There may be more than one for some actions, and some benefits may be used more than once.

Action timetable	**Benefits (numbers)**
Feral Water Buffalo allowed to graze in Kakadu up until 1980s	
Feral Water Buffalo removed from Kakadu	
Mudja grass grew and choked off wetland	none
CSIRO scientists consulted with Indigenous people and set up the burn timetable	
May to August burn-off. Paperbark forests and other woodland around wetlands burnt.	
September to December burn off. Wetland. Dead Mudja grass burnt and living grass killed.	
Later in season (before December 'big wet' arrives). Tuft burn off to get rid of dry grass tufts killed in earlier burns	
Wet season. There are now open water areas and ground available for growth.	

Benefits

1 Mudja kept down
2 wildlife recovers
3 ground softens
4 learn about burning
5 controls spread of burn-off
6 makes space for native plants
7 water lilies, wild rice and spike rushes grow
8 turtles and magpie geese thrive

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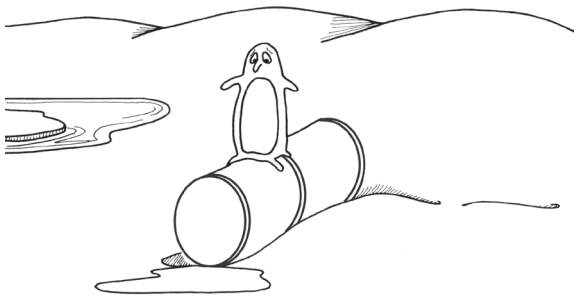
Antarctica



☞ **1** Replace the missing vowels (a, e, i, o, u) in the following passage.

The A__str__l__ __n Ant__rct__c D__v__s__ __n says that hum__n imp__cts on Antarct__ca include h__nt__ng and d__m__g__ng animals and pl__nts, c__nt__m__n__t__ng soils, disch__rging sew__ge into the s__a, and leaving r__bbish, fuel dr__ms, tr__cks and monuments made of st__n__s, called cairns. Eff__rts are being made to cl__ __n up the area.

2 Many different things are being done to reduce human impacts on Antarctica. Number the following actions from most important (1) to least important (14).

Environmental action	Importance	Environmental action	Importance
Move huskies away from Antarctica		Stop whaling	
Remove all old rubbish dumps		Adopt an international treaty to prevent overfishing	
Remove shipwrecks		Clean up contaminated soils	
Treat sewage and clean up old mess		Break down cairns	
Prevent new species from being introduced		Make rules about how close people can come to penguins, seals and other wildlife	
Stop krill fishing		Prevent mining	
Study the local environment in detail		Reduce global warming	

3 Explain why you chose your number 1 action in the table. _____

_____

_____

_____

_____

_____

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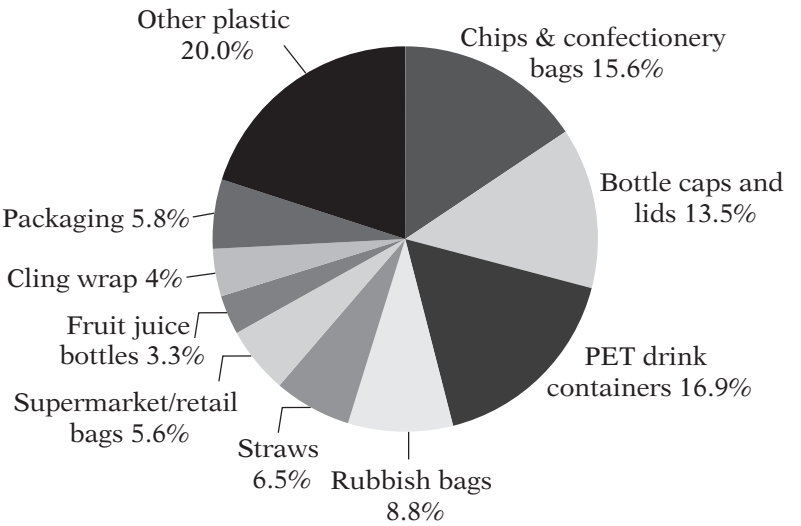
Cleaning up Australia

Every Clean Up Australia Day, groups of volunteers around Australia choose a local area to clean up. In 2010 more than 90 000 plastic items were picked up. Those items were made of 30 different types of plastic. The top ten types are shown in the pie chart below.

Breakdown of plastic items



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☞ **1** Choose the correct option from each pair in brackets.

The chart shows only the [wooden/plastic] items found.

There are [three/two] types of items from drink bottles and containers; four if you count straws.

Different kinds of bags make up [8.8%/30.0%] of the plastic items found.

If [drink bottles and containers/chip bags] were recycled, that alone would reduce the collected items by [4%/20.2%], without counting lids or straws.

2 Two suggestions have been made to reduce rubbish. One is to have people pay a deposit on drink bottles, which would be refunded when the bottles are recycled. The other is to charge for shopping bags. Which do you think would have the greatest effect? Why?

_____

_____

3 Can you suggest three other ways to keep Australia clean?

_____

_____

_____

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Crop technology

Huge amounts of wheat are grown in Australia. Crops grown close together can suffer from outbreaks of disease. Scientists have identified a **gene** that makes a protein called **Lr34 transporter protein**. This protein protects wheat plants against three diseases – leaf rust, stripe rust and powdery mildew – that can severely damage wheat crops. Now that the gene sequence of the protein has been identified, it can be implanted in wheat plants and bred widely.

☞ **1** Use the word bank to complete the following passage.

Word bank

growing benefit protects living cooperating
future passes Millions disease gene

Plants are _____ things. They can get sick. _____ of plants grow together in a farming area such as a wheat- _____ area. If one plant is sick and _____ the disease on, a whole year's crop can be wiped out. If a few plants do not get sick, this may mean they have a gene that _____ them.

If a whole crop is planted with that protecting _____, the crop is safe from the _____. The Lr34 gene will protect _____ crops against three fungus infections. Earlier attempts to prevent disease have had short term results, but the _____ of Lr34 will last for a long time.

Australia's CSIRO scientists are _____ with others around the world to improve crops. A new type of rice plant is being developed which will produce much more food, for example.

2 Write a sentence explaining how the problem of world hunger could be partly solved by the new rice plant that is being developed. _____

_____

_____

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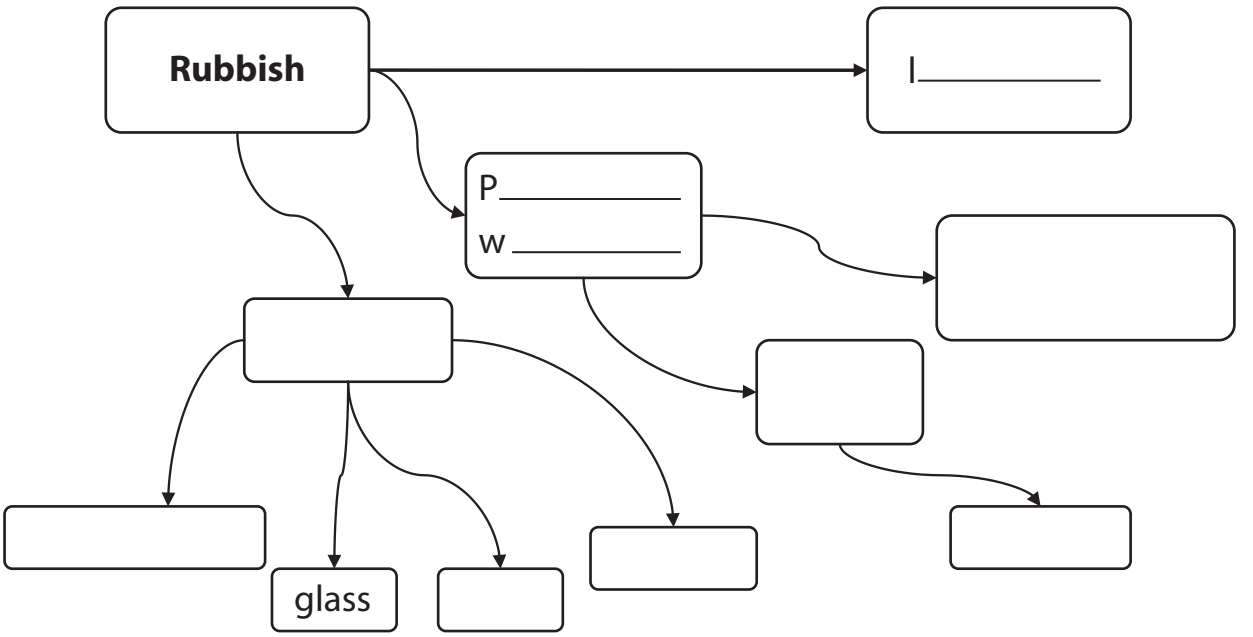
Waste recycling

Humans produce a huge amount of rubbish. Think about all the packaging that is used for the items that we buy every day. We need to think carefully about how to dispose of that **waste**. Dumping rubbish causes serious pollution risks and wastes resources that could be re-used. Metals, glass and many kinds of plastics can be **recycled**, which saves the raw materials and the energy needed to process waste.

☞ **1** Use the word bank to complete the labels on the diagram below.

Word bank

landfill compost plastics aluminium food scraps steel
Recycle Plant waste green waste collection



2 List five materials that you recycle.

_____ _____

_____ _____

_____

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Teacher's notes – Microscopes and cells

Achieve!

Objectives

- To remember that living things are made from cells
- To learn that development of imaging technology was slow and gradual
- To understand that science and medicine improve with new technologies
- To learn how to interpret tables of information

Prior knowledge

Some knowledge of cells may be helpful, but is not essential.

Background

Knowledge depends on seeing or otherwise observing. Until cells became visible we didn't know they existed. It wasn't until lens technology was developed that we discovered cells. As the technology improved so did our knowledge of cells. Now we are on the verge of finding a stem (origin) cell for a tissue and repairing that tissue.

Starter activity

Give students practice in forming and testing hypotheses. Place a mystery object such as a vase or drink bottle filled with pond water under a tea towel. Allow the students to ask questions about the mystery object. As more evidence is revealed, students should form new and improved hypotheses. Students may end up examining the pond water with microscopes or hand lenses. Use of a microscope, hand lens or a photograph of cells taken through a microscope will assist.

Ask if any students have had experience of an X-ray to prompt a starter discussion about medical technology.

Activity sheets

The Activity sheet 'Single lens to compound microscope' presents a simple history of people working to improve the microscope. Students must choose the right word from each pair in order to complete a passage summarising developments.

The 'Improving cell theory' Activity sheet demonstrates how scientists are constantly revising and refining theories. Students match sentence halves to show that they understand how hypotheses are modified with new evidence.

The 'New imaging technology' Activity sheet aims to familiarise students with technical words and abbreviations, while also introducing some of the results of human endeavour in medical technology. Students match abbreviations with full names and answer comprehension questions.

'Human and tree cells' presents a crossword that examines the differences between plant and animal cells. Students are provided with a word bank to prompt them.

The Activity sheet 'Stem cells' describes current biological research into the potential use of stem cells. Students complete a 'fill in the vowel' activity before answering comprehension questions to conclude the activity.

Extension ideas

Hold a class debate about the use of embryonic stem cells, exploring the ethical issues as well as the potential benefits. Encourage students to research Christopher Reeve and the work of the Christopher Reeve Paralysis Foundation. The following websites provide some additional background information on this topic.

- http://stemcells.nih.gov/info/basics
- www.christopherreeve.org

Selected answers

Human and tree cells Across: **3** oxygen, **4** cellulose, **6** sugar, **7** photosynthesis, **10** muscle, **12** carbon dioxide, **13** wall, **14** cellular respiration; Down: **1** cell, **2** bones, **5** light, **8** tree, **9** human, **11** cartilage.

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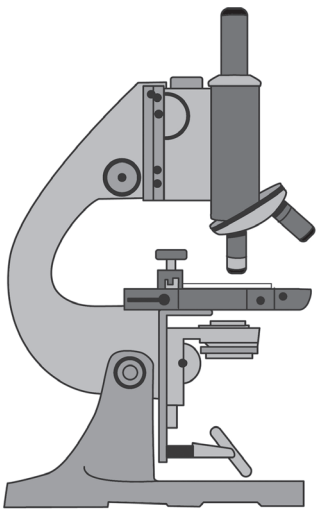
Single lens to compound microscope

The first microscope was used in the 1500s. Over time, microscope technology has steadily improved. As the technology improved so did knowledge about **microbes** – living things that are too small to see with the naked eye.



Compound microscope

- At first, **specimens** were magnified with single lenses. Some of the scientists involved at the time were Sacharias Jansen in around 1595, Robert Hooke in 1665 and Antonj van Leeuwenhoek in 1653.
- Hooke and Jansen later used two lenses in a row, sometimes inside a barrel. These were the first compound microscopes.
- Hooke was the first to try shining a light directly onto the specimen.
- Van Leeuwenhoek was first to see pond life protozoa, spirogyra and sperm of many animals.
- Hooke made very thin slices of cork that let light through, and saw little compartments, which he named **cells**.

☞ **1** Choose the correct word or phrase in each pair.

Microscopes

The lens of a [microscope/camera] lets only a tiny amount of [light/water] through. Light can be blocked by using too much of a specimen or a slice from a specimen that is too [wide/thick]. Robert Hooke cut thin slices of cork that let light pass through so he could see the cells. He also worked out a way to light up the specimen with a small reflecting [lamp/light globe].

Antonj van Leeuwenhoek used tiny lenses made by melting and pulling glass into bead shapes. He sometimes used [one/two] lenses – one in each hand – to get a [clearer/brighter] view. This was like a [compound/simple] microscope without the barrel. Improvements in lenses and microscopes over [tens/hundreds] of years show how the human endeavours of science and technology develop alongside each other.

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Improving cell theory

Scientists improve their theories as new evidence is collected. The idea of **cells** changed over a long time as more evidence was found.

- 1665 – Robert Hooke first saw and named cells. Possibly the first theory would have begun as: *There are some cells in living things.*
- 1839 – Matthias Schlieden and Theodor Schwann revised the original idea to read: *All living things are made from cells.*
- Soon after, Schlieden and Schwann decided that the theory should be revised to include a reference to the materials that cells make, such as bone matrix.
- 1858 – Rudolf Virchow added the idea that all cells come from other cells. He is now believed to have copied this idea from Robert Remak.

The theory was then very close to the definition we use today:

All living things are made from cells and their products. All cells come from other cells.

Science is open to change, but the changes will only be accepted if they are very strongly supported and not able to be disproved.

☞ **1** Match these sentence halves to summarise the ideas above.

Living things are made of	lot of cell product around every cell.
Hooke used his improved microscope to observe	but become famous by understanding the importance of what they have seen.
The matrix is the material	cells in cork and to name them 'cells'.
Schlieden and Schwann's theory was improved	cells and cell products. Cells always come from other cells.
Some tissues, such as bone, have a	that surrounds the cell in tough tissues.
Scientists sometimes see things others have seen before	there is evidence for the change and it has not been disproved.
Scientific theories only change when	after many observations and communications between scientists.

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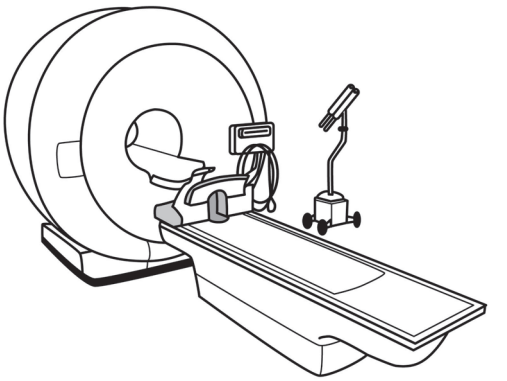
New imaging technology

Medical technology has improved greatly over time, and doctors now have many ways of looking inside patients without cutting them open.

X-rays don't pass through bone, so they can make an image of the bone like a shadow on a film or screen. X-rays can pass through soft tissue in varying amounts, making shadows that are weaker than bone images.

A **CAT** scan is a computer controlled series of X-rays that shows an area of the body in a series of 'slices'. This allows doctors to see tumours or injuries.



Magnetic Resonance Imaging machine

Ultrasound vibrations are used to make an image of an internal organ such as a heart, or of an unborn baby. **Magnetic Resonance Imaging** (MRI) uses large magnets and radio waves to make a three-dimensional image.

☞ **1** Draw a line matching each technique with its abbreviation.

MRI	Diffuse Optical Imaging – uses near infrared light
CAT	Event-Related Optical Signal – uses infrared light through optical fibres
MRE	Computed Axial Tomography – uses X-rays controlled by a computer
DOI	MagnetoEncephaloGraphy – uses magnetic field of the brain
EROS	Image-directed Colour Doppler Ultrasound – uses ultrasound vibrations
EEG	ElectroCardioGraphy – uses electric signals of the heartbeat
ECG	ElectroEncephaloGraphy – uses the brain's own electric signals
ICDU	Magnetic Resonance Elastography – uses huge magnets and radio waves
MEG	Magnetic Resonance Imaging – uses huge magnets and radio waves

2 Which technique could be used to show a brain in a series of slices?

Abbreviation: _____. Full name: _____.

3 To record the electrical pulse of the heart a doctor would use:

Abbreviation: _____. Full name: _____.

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Human and tree cells

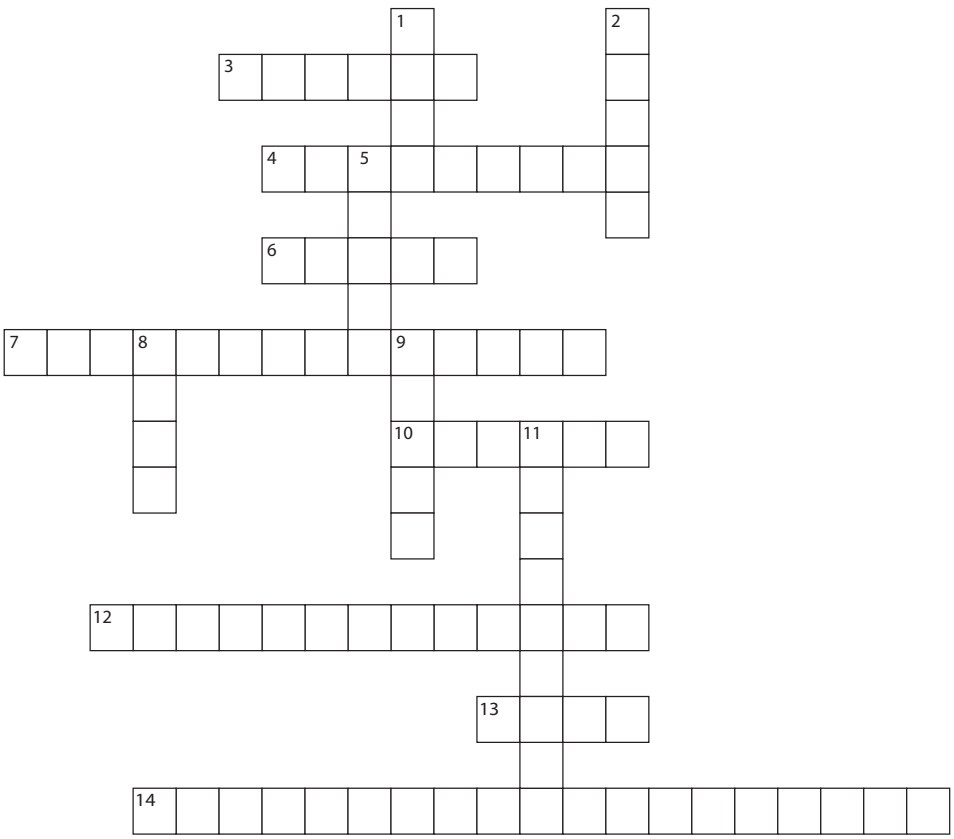
Tree cells and human cells have many similarities – for example, both human and tree cells use **cellular respiration** for energy. However, there are also some big differences.

- Green tree cells produce their own food through **photosynthesis**. They use light to make sugar from carbon dioxide and water.
- Plant cell walls are made from a substance called **cellulose**.
- Some human cells make products to build strong tissue such as bone or cartilage.

☞ **1** Use the word bank to complete the crossword puzzle.

Word bank

oxygen muscle light photosynthesis tree wall bones cellular respiration cartilage human carbon dioxide sugar cellulose cell



Across

3. Trees and humans need this
4. Material of plant cell wall
6. Useful for energy
7. The process by which green plants produce food.
10. Helps move human arms and legs
12. Plants use it with light and water to make sugar
13. Strong part of plant cell
14. Trees and humans do this all of the time

Down

1. Tiny part of a human or tree
2. Human skeleton
5. Green plant cells can use it for energy
8. A strong, woody plant
9. Person
11. Tough tissue in nose and ear

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Stem cells

☞ **1** Replace the missing vowels (a, e, i, o, u) in the first two paragraphs.

A single f__rt__l__s__d egg divides into __v__ry d__ff__r__nt type of c__ll that exists in an adult. V__ry early in the lif__ of the div__ding egg there are **stem cells**, which can d__v__d__ to change into specialist cells such as m__scle cells or blo__d cells.

There are two types of stem cells – **embryo** stem cells and **adult** stem cells. Embryo st__m c__lls can gr__w into any type of c__ll. Ad__lt st__m cells __nly become the type of cell of the t__ss__ __ they live in. Some adult stem c__lls can be made to b__h__ve like __mbry__ cells to pr__d__ce any type of cell. These are c__ll__d 'induced pluripotent' st__m c__lls.

Scientists hope to use stem cells to grow new cells in damaged tissue. This could help to cure diseases. For example, stem cells could be used to grow new pancreas cells in diabetics, heart cells in heart attack sufferers, and so on. High priority will be given to treating spinal injuries and disorders where cells stop working properly. Doctors will need to ensure that injected stem cells are not rejected by the body, which could see the new cells as invaders, and that the stem cells don't cause new problems.

☞ **2** What do fertilised egg cells divide into? _____

3 Muscle and blood cells are examples of what types of cells?

_____

4 What can adult 'induced pluripotent' stem cells do?

_____

5 What is it that scientists hope will grow new cells in damaged tissue?

_____

6 Where would a diabetic have stem cells implanted? _____

7 Why do you think spinal injuries will be a high priority? _____

_____

8 What are two of the problems that doctors have to solve with stem cell treatments?

_____

_____

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Teacher's notes – Transplants, grafts and ethics

Achieve!

Objectives

- To be aware that Australian scientists are prominent on the world stage
- To understand that development of new ideas and techniques needs to be ethical
- To know how to interpret and construct flowcharts
- To learn that donation of blood, tissues or organs is a way individual people can save lives
- To understand that science is a method of thinking and operating, not a belief system

Prior knowledge

Students need to have some knowledge of flowcharts before attempting to construct one in 'Spray-on skin'. Some knowledge about grafts and transplants may be useful.

Background

Cells of humans and animals have recognition signals on their cell membranes. If they are transplanted or transfused into a different individual they will be recognised as an invader and attacked by the body's defences. Blood has to be very carefully typed. Transplant recipients must take drugs to suppress their defences. The best option is to use a person's own cells where possible. Transplants and donations of tissues or organs receive frequent media exposure. The activities in this unit examine different aspects of this practice. The ethics of transplanting must be carefully controlled to avoid scenarios such as organ smuggling or unethical breeding of 'spare part' donors.

Starter activity

Find a recent media article about a graft or transplant and use it to start a class discussion. Have students explore the advantages of transplant technology, but also explore the ethical issues surrounding the topic. Students' anecdotes about family experiences can enrich the discussion, but it is important to be wary of upsetting students who have suffered recent bereavements or who have family members who are ill.

Activity sheets

The Activity sheet 'Spray-on skin' outlines the achievements of Professor Fiona Wood and Marie Stoner. Students complete a flowchart of the order of grafting events using a word bank of phrases.

In 'The gift of life', students explore the topic of organ donation and transplants. They fill in missing vowels and then choose the correct word from brackets to consolidate their understanding of tissue or organ donation.

The Activity sheet 'Breeding humans?' asks students to categorise a list of hypothetical acts as right, wrong or illegal. This helps students understand the importance of ethics in medicine, and that some actions may be illegal.

In the Activity sheet 'Do you believe in science?' students are asked to consider hypotheses that are supported and unsupported. The activity encourages critical thinking and gives students practice in reaching a considered conclusion based on the available evidence. This Activity sheet could be used with the 'Improving cell theory' Activity sheet from Unit 3, which outlines the scientific steps taken in developing cell theory.

The 'Transfusion and cell recognition' Activity sheet explains the importance of blood type compatibility. Students complete a cloze passage, then identify a nonsense sentence to be excluded.

Extension ideas

Any of these activities could be used as the basis for a class debate. Debating topics such as alien pyramid builders or using humans as organ banks can help demonstrate the importance of ethics and the need for evidence. The following websites can provide additional background information:

- www.mccomb.org.au
- www.transfusion.com.au
- www.donateblood.com.au
- www.redcross.org.au

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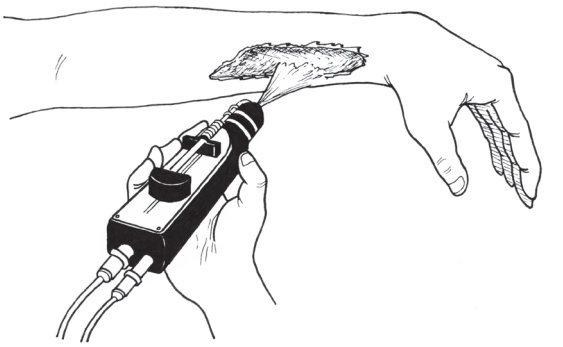
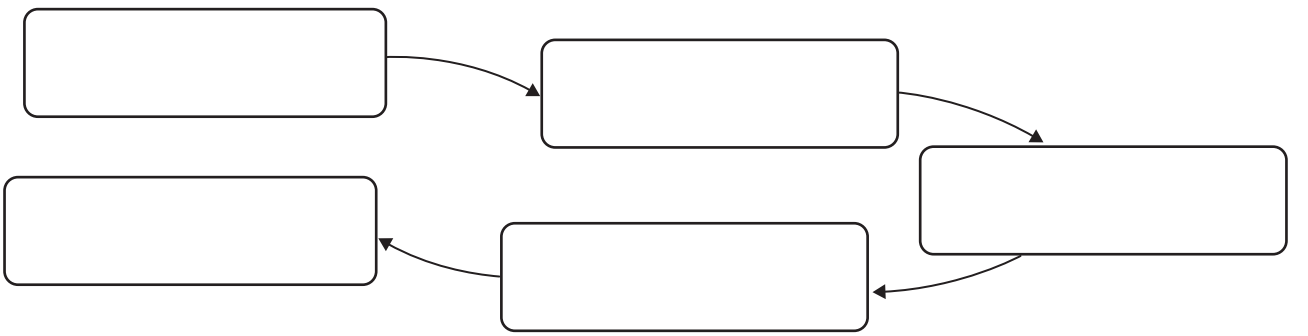
Spray-on skin

Spray-on skin is an invention used to help patients regrow new skin after suffering severe burns. It was developed in Australia by Professor Fiona Wood, a plastic surgeon, and Marie Stoner, a scientist. To make spray-on skin, skin cells are taken from the patient. They are mixed with liquid food and grown for five days, then sprayed onto the burn using a special cell sprayer. New skin grows very quickly. Spray-on skin was used to treat victims of the 2002 Bali bombings.



☞ **1** Use the labels in the word bank to complete the flowchart.



Word bank

mixture sprayed on burn

skin cells grown in culture

skin taken from unburnt area

new areas of skin grow on burn

skin cells in sprayer

2 Describe how this invention helps burns victims. _____

_____

_____

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The gift of life

☞ **1** Read the passage carefully and insert the missing vowels (a, e, i, o, u).

M__ny Australians s__ffer from a seri__us disease that could be relieved by a tr__nspl__nt. Some people choose to d__n__t__ their organs after they die so that they can be used for tr__nspl__nts. P__ __pl__ who want to donate their __rg__ns can join the A__stralian Organ Donor R__g__st__r. If a p__rs__n is in an __cc__d__nt, their n__xt of kin may have to d__c__d__ whether to donate that person's org__ns or not. J__ __n__ng the Register makes a person's w__sh__s clear.

Organs fr__m one person could save many l__v__s. Organs that can be d__n__t__d include the h__ __rt, lung, p__ncr__ __s, kidney and liver. Tiss__es such as corneas, skin, heart v__lv__s and pericardium can also be d__n__t__d.

2 Choose the correct word from each pair in brackets.

One tissue – blood – is [wasted/donated] and used in thousands of litres every day. The blood donor's [body/family] quickly makes new blood to replace the donation. The blood can be used to help many people, such as cancer patients, accident victims and people having surgery.

Most other organs and tissues are only available from people who have [died/spare ones]. However, a living person can donate bone marrow, a kidney or a part of a liver or pancreas.

There are national registers of [patients/hospitals] who are waiting for [transplants/transport]. Organ recipients are chosen from the national registers according to [tissue/napkin] matching, their size compatibility, how long they've been waiting, their medical condition at the time, urgency, and whether the organ can be supplied to them in time. They are tested for [compatibility/intelligence] with the new organs to avoid rejection – the [biggest/least] possible problem. Once the new organs are transplanted, the patients are [sprayed/treated] with anti-rejection medicine for the rest of their lives. Only those who receive a donation from their [identical twin/friend] would not need the anti-rejection medicine.

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Breeding humans?

The biggest danger with organ and tissue transplants is **rejection**. A patient's body will see a new organ as an invader and try to get rid of it. Transplant recipients must take anti-rejection drugs to stop this from happening. The drugs are called **immunosuppressants**, because their job is to suppress the **immune system**. Rejection is less likely to occur if the organ comes from a close relative. Identical twins are the best match, and a brother or sister may be the next best.

Imagine you could grow a copy of yourself – then you would have the perfect match if you ever needed a transplant! But would that be fair on the other 'you', who only existed to donate organs? Acts like that could be illegal.

Apart from laws, doctors also have rules about treatment called **medical ethics**. Government health departments and hospitals have ethics committees. The committees make rules that say which medical actions are right and wrong.

☞ **1** Decide whether you think each of these examples is right, wrong or illegal. Write 'R' for right, 'W' for wrong and 'I' for illegal.

- Grow a person's cells and then inject them back into the same person. _____
- Accept a donation of a kidney from a brother or sister. _____
- Accept a culture of bone marrow cells that come from another person. _____
- Accept a heart from a person killed in an accident. _____
- A heavy smoker receives a lung transplant from a dead accident victim. _____
- Have a second child so they can donate a kidney to an existing child. _____
- Purchase a kidney from a poor person in a developing country. _____
- Buy a transplant heart from a secret supplier in another country. _____

2 Choose one of the above examples and write a paragraph in your workbook explaining your opinion about whether it is right, wrong or illegal.

3 Who do you think should make final decisions in cases like these?

_____

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Do you believe in science?

It is difficult to believe in science because the whole point of science is not to believe. Every fact or law of science is based on what we call a **hypothesis**. A hypothesis is a theory. It can be **accepted as true** for two reasons:

- there is a large amount of supporting evidence
- it has not been **disproven**.

When both of these reasons apply, it has been accepted as science. People sometimes use the second point alone to manipulate information. For example:

> *Aliens built the pyramids. This has been scientifically proven because it has not been disproven.*

What evidence is there? A spacecraft landing zone beside the Sphinx? Be wary of the words 'scientifically proven'. Science has strongly supported hypotheses, not proof. New evidence may arrive any day to improve or disprove a hypothesis. Most famous scientists have had their brilliant hypotheses changed at some point.

☞ **1** Underline the statements that you think could be called **unsupported hypotheses**.

- My old house has ghosts.
- Crop circles are made by landing alien spacecraft.
- A car that hits a tree at high speed is likely to be badly smashed.
- Smoking won't hurt you. My granddad smoked and he lived to be 91.
- A parachutist falls slowly enough to reach the ground safely.
- My grandma tells fortunes. She said I'd meet a handsome, dark-haired boy and I did.
- The forecast says it will rain tomorrow.

2 Write two supported statements and two unsupported statements.

Supported statements:

- _____
- _____

Unsupported statements:

- _____
- _____

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Transfusion and cell recognition

'Jack is bleeding badly. He needs a transfusion.'

'Oh, he can have some of my blood.'

But Jack can't, unless his blood type is compatible with the donor's. The human body recognises blood cells. The **immune system** recognises all cells as either 'my cells' or 'invaders' and it attacks the invaders. **Cell recognition** is important for blood transfusions but also for organ and tissue transplants.

☞ **1** Use the word bank to complete the sentences below. One sentence is completely wrong – cross that one out.

Word bank

positive paramedics hurts good stitches anaesthetic chainsaw sample blood nurse

Casualty

Annie has sliced her arm open with a _____ while cutting firewood. At casualty, the _____ tell the doctor that she has lost a lot of _____. The doctor asks the nurse to get a blood _____ quickly to the laboratory. The trainee _____ puts *Iron Maiden* on the stereo because iron is _____ for the blood. The nurse comes back soon with a 600 ml bag of O _____ blood because the test showed that Annie was O positive. Two nurses quickly set up a transfusion of blood while the doctor _____ the wound.

'I bet that _____,' said the trainee.

'No,' said the doctor, 'I gave her an _____.'

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Teacher's notes – The battle against disease

Achieve!

Objectives

- To understand that nanotechnology, although in its early stages, has amazing potential
- To learn about the work of Australian scientists
- To be aware that citizens need to read media reports critically
- To know that donated blood has many uses

Prior knowledge

Students need to understand that sickness is part of everyone's lives and science and medicine are constantly seeking ways to overcome it.

Background

Medical treatment improves because great scientists make it possible. Excellent media coverage can help science, but bad or overly selective coverage can damage it. Nanotechnology is one of the exciting possibilities that could either be damaged or encouraged by media reporting, depending on how balanced the reporting is.

Starter activity

Hold a class discussion about developments in medicine over time. Ask students if they can name any diseases that were problematic in the past but are now rarely seen, if at all. What kind of developments led to the treatment of these diseases? How might changes in people's attitudes have contributed to these developments? Emphasise the great contribution that has been made by Australian scientists, but also point out how individuals contribute to change.

Activity sheets

The Activity sheet 'Donating blood' presents a flowchart showing what happens to donated blood. Students select the correct word from pairs to demonstrate their understanding of the process.

The 'Nanotechnology' Activity sheet introduces students to nanotechnology and its potential. Students replace missing vowels, then match sentence halves to consolidate their understanding.

'Immunisation and the media' teaches students to read with care and to be aware of context. An air crash may lead to headlines stating '200 killed' but around 3 million people fly every day. Victims of immunisation are publicised but the billions who are saved by it are not. Students match answers to questions as a way of putting critical thinking into practice.

The Activity sheet 'A great Australian scientist' outlines the work of Sir Frank MacFarlane Burnet. Students complete a cloze activity to reinforce the significance of his research.

The 'Fred Hollows' Activity sheet examines the link between biology and medicine, shown in the work of Fred Hollows. Students choose from alternative words to complete sentences.

Extension ideas

- Students could research an Australian scientist and design a poster or PowerPoint presentation outlining their work and its significance.
- Students could analyse a media article, looking for accuracy or sensationalism to follow up the 'Immunisation and the media' Activity sheet.

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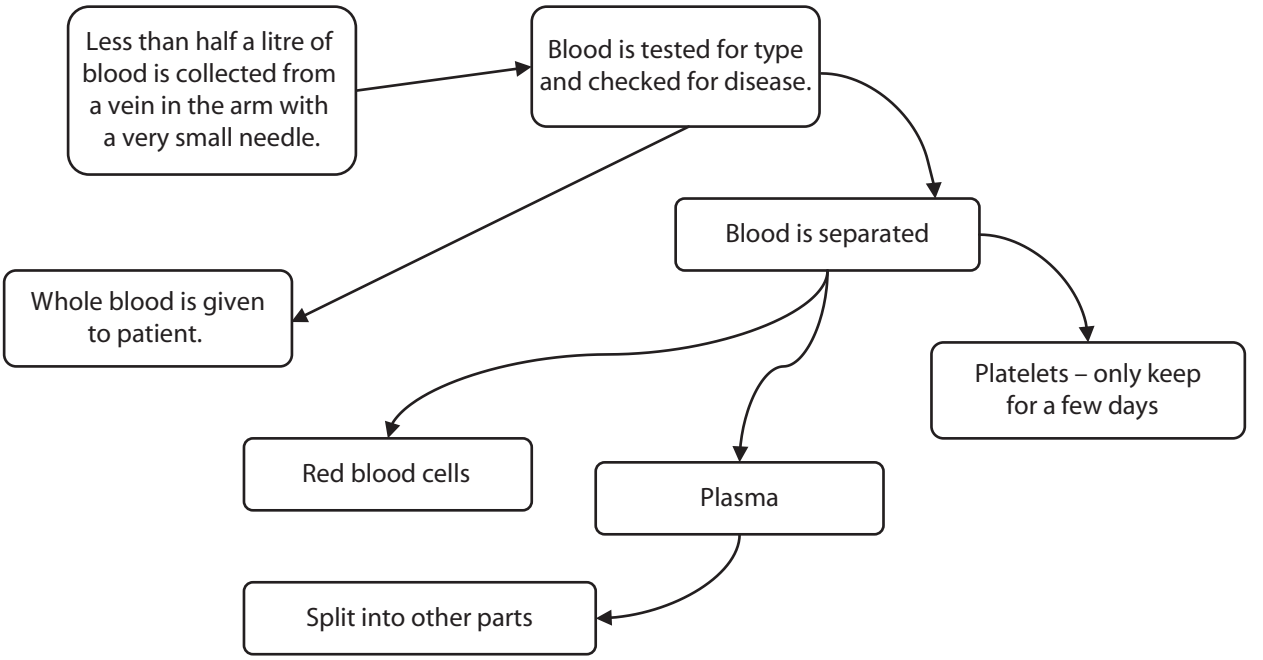
Donating blood

The Australian Red Cross Blood Service collects and distributes blood and blood components. One in three people will need blood at some point in their lives, but only one in thirty people donate it.

What happens to donated blood?



☞ **1** Fill in the missing words and choose the correct word from each pair to complete the passage below.

After the blood [colour/type] is checked, donated blood can be given whole to a patient. However, often the blood is separated into its parts – p_____, p_____ and r___ b_____ c_____. Doctors use the part that will be most effective and [wasteful/efficient] in helping the patient. **Whole blood** is not used if only [platelets/aspirin] are needed, for example. **Red blood cells** are used for anaemia and for high blood [gain/loss] in surgery. **Platelets** help clot wounds, but are often destroyed by chemotherapy. They are transfused into [cancer/flu] patients or used in high blood loss surgery. Platelets only keep for a [hundred/few] days. **Plasma** can be used after injury causing [severe/small] blood loss, and can be split into many parts for the best use, such as with or without different clotting factors.

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Nanotechnology

A **nanoparticle** is tiny. It is a **molecule** or part molecule and is sometimes called a nanomolecule. They can travel easily through blood vessels and enter some cells. Scientists are trying to find ways to use these tiny chemical particles in medicine. Possible uses could be:

- coating a medicine with nanoparticles so it goes only to the organ where it is needed
- attaching nanoparticles to a medicine so it can cross through a body barrier, such as from blood into the brain
- to give small doses of medicine that isn't wasted in other body parts because it only goes where it is needed
- protecting a vaccine so it can get into the cells where it can work
- delivering dangerous chemotherapy drugs directly to the cancer cells only
- finding damaged cells and repairing them
- attaching nanoparticles to a tumour to make it visible to imaging machines
- making purer medicines to reduce side effects caused by impurities.

Nanotechnology requires many years more research. Because the nanoparticles are so small and not enough is known about them yet, there are possible dangers. Will they stay in the body long after their use is over? Will they stay unchanged in the body or could they change and cause problems? Will they get into the environment and cause pollution problems? These possible dangers are studied in **nanotoxicology**.

☞ **1** Replace the missing vowels, then join the sentence halves to make complete sentences.

Nan__p__rt__cl__s may find	against potential d__ng__rs of nanoparticles.
Nanoparticles may __tt__ch to	to make s__r__ they go to their t__rg__t.
N__n__m__l__c__l__s may coat dr__gs	t__m__ __rs and help with early d__ __gn__s__s.
Nanoparticles may __ll__w us to use	d__m__g__d cells and f__x them.
N__n__t__x__c__l__gy is trying to g__ __rd	much smaller am__ __nts of m__d__c__n__s.

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Immunisation and the media

We rely on the media to provide us with information. However, we need to be aware of how the information is being presented, and whether all the information is being reported. How often have you seen a news story saying that everything is all right? Good news stories are not very dramatic, so we don't see them in the news very often.

The topic of immunisation is one that people have different opinions about. Immunisation can save millions of lives by preventing the spread of disease. However, there have also been stories of immunisation having terrible consequences, such as brain damage, in children. These stories may sometimes be reported more widely than the success stories.

It is important to study the news and history while also asking questions such as:

- What is the risk of the vaccine?
- What is the risk of the disease it is trying to stop?
- Is the risk just to my child or others as well?

Reading the news in this way allows individuals to form a considered opinion.

☞ **1** A local headline read 'Children ill after National Vaccination Program'. As a result, worried parents asked questions at the local medical clinic. Match the answers given to each question below.

Questions	Answers
How sick are the children?	Children used to become extremely ill and many died.
Are the sick children local or from all over Australia?	No, it is my professional opinion that it is the common cold.
Are children who didn't have the vaccination also sick?	Several unvaccinated children are also sick.
How serious was the illness being immunised against before wide-spread vaccinations began?	Several local children are sick but so are children from a wider area.
Is it the doctor's opinion that the vaccination made the children sick?	The children are only slightly ill.

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A great Australian scientist

Imagine you are walking along a corridor. It is 1965. You can smell chickens. The Walter and Eliza Hall Institute of Medical Research is not where you expect to smell a hen house! You follow your nose and walk into a neat, spotless laboratory. A gentleman with a starched white lab coat over his suit stands at the bench with an egg in his hand.

You have stumbled upon Sir Frank Macfarlane Burnet (1899–1985) who won a Nobel Prize for his work in **immunology**, the study of disease resistance, in 1960. Mac, as his colleagues knew him, was Australia's most celebrated scientist. He researched dozens of topics in immunology and **virology** (the study of viruses) as well as other areas, including ageing.

Burnet used egg membranes to grow **viruses**, including the influenza virus, which is related to bird viruses. While he was a great thinker, he was also a practical scientist who spent half of each day in his laboratory conducting experiments.

☞ **1** Use the word bank to complete the following passage.

Word bank

viruses immunology experiments Prize important Nobel disease resistance sciences Walter ageing

The Nobel _____ is a prestigious award given for peace, literature and various _____. Burnet's work was mostly in virology but he won the _____ Prize for an advance in _____. Chicken eggs were used to grow viruses and Burnet carried out very _____ work on influenza. Although he was very busy as Director of the _____ and Eliza Hall Institute, he still did _____ every day. His contribution to our understanding of disease, immunity and _____ was outstanding.

Virologists study _____ and immunologists study _____ _____.

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Fred Hollows

Although he was originally a New Zealander, Fred Hollows is one of Australia's most famous scientists – he was even named Australian of the Year in 1990.

In the 1960s Hollows helped to set up the first Aboriginal Medical Service. By 2005, there were more than 60 of them around Australia. As an eye specialist he was horrified to learn that many Indigenous Australians were blind from **trachoma** – a preventable and treatable eye disease. Hollows did many trachoma repair operations himself and started up the National Trachoma and Eye Health Program to inform Australians, to raise funds and to carry out operations.

Hollows didn't stop there. He and his teams visited many poor countries to carry out operations; they even set up a lens factory in desperately poor Eritrea. When he became terminally sick with cancer he set up the Fred Hollows Foundation, an organisation that works to prevent avoidable blindness and to improve Indigenous Australians' health. Since 1992 the Foundation has worked in more than 30 countries in Asia, Africa, Australia and the Pacific. The one millionth intraocular lens (a device used to treat cataracts, which cause blindness) was inserted in 2003.

Fred Hollows helped millions of people, and his legacy continues to help millions more.

☞ **1** Circle the correct word in each pair.

Fred Hollows treated [eye/leg] diseases. He argued [strongly/weakly] for teams to work with him to treat trachoma and other eye [diseases/glasses]. He [raised/wasted] money from people, governments and companies to [support/delay] his efforts. Many [nasty/harmless] eye diseases happen in places where people are very [poor/wealthy]. Hollows and his teams went to these places to help hundreds of thousands of people. He was a great Australian and a great [New Zealander/Canadian].

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Teacher's notes – Separation and waste

Achieve!

Objectives

- To understand why recycling and waste disposal are critical to society's sustainability
- To learn how automated separation systems carry out recycling at council depots
- To know that citizens need to take responsibility for recycling at home
- To learn that separation techniques and biological treatment make sewage safe
- To learn how to interpret and construct flowcharts

Prior knowledge

Students should be familiar with household rubbish bins and recycling bins, and should understand that kitchen and toilet waste is kept separate from other waste.

Background

The separation techniques outlined in this unit are mimicked on a huge industrial scale by council recycling depots and sewerage farms.

Starter activity

The Activity sheets 'Different types of water' from Unit 1 and 'Waste recycling' from Unit 2 are good preliminary activities to this unit. As an additional starter activity, list various waste materials on the board, such as plastic bags, glass bottles, cardboard boxes, disposable nappies, plastic bottles, glass jars, broken window glass, aluminium cans, steel cans, dirty tissues, detergent bottles, light globes, ice-cream containers, vegetable scraps, meat scraps, yoghurt containers and ceramics. As a class, decide whether each kind of waste should go in the recycling bin or the garbage bin.

Resource and Activity sheets

The Resource sheet 'Recycling separators' presents a diagram of an entire recycling separation system. This information relates to the following Activity sheets, which explain the stages of the process that takes place after the recycling bin is collected.

The Activity sheet 'Disc screen and magnetic roller' looks at separation of paper and steel from other contents. Students complete a cloze activity and comprehension questions to analyse the process.

The 'Air classifier' explains how air blasts are used to separate dense glass from aluminium and plastics. Students choose from multiple options to complete sentences, and respond to a comprehension question.

The Activity sheet 'Optical sorting' presents a diagram illustrating how both glass and plastic bottles are sorted. Students complete a cloze task and comprehension questions to demonstrate their understanding of the process.

'Sewerage' outlines the separation and treatment processes involved in sewerage. Students match sentence halves about sewage treatment.

Extension ideas

Give students a variety of recyclable objects and have them construct a flowchart showing how the objects could be separated.

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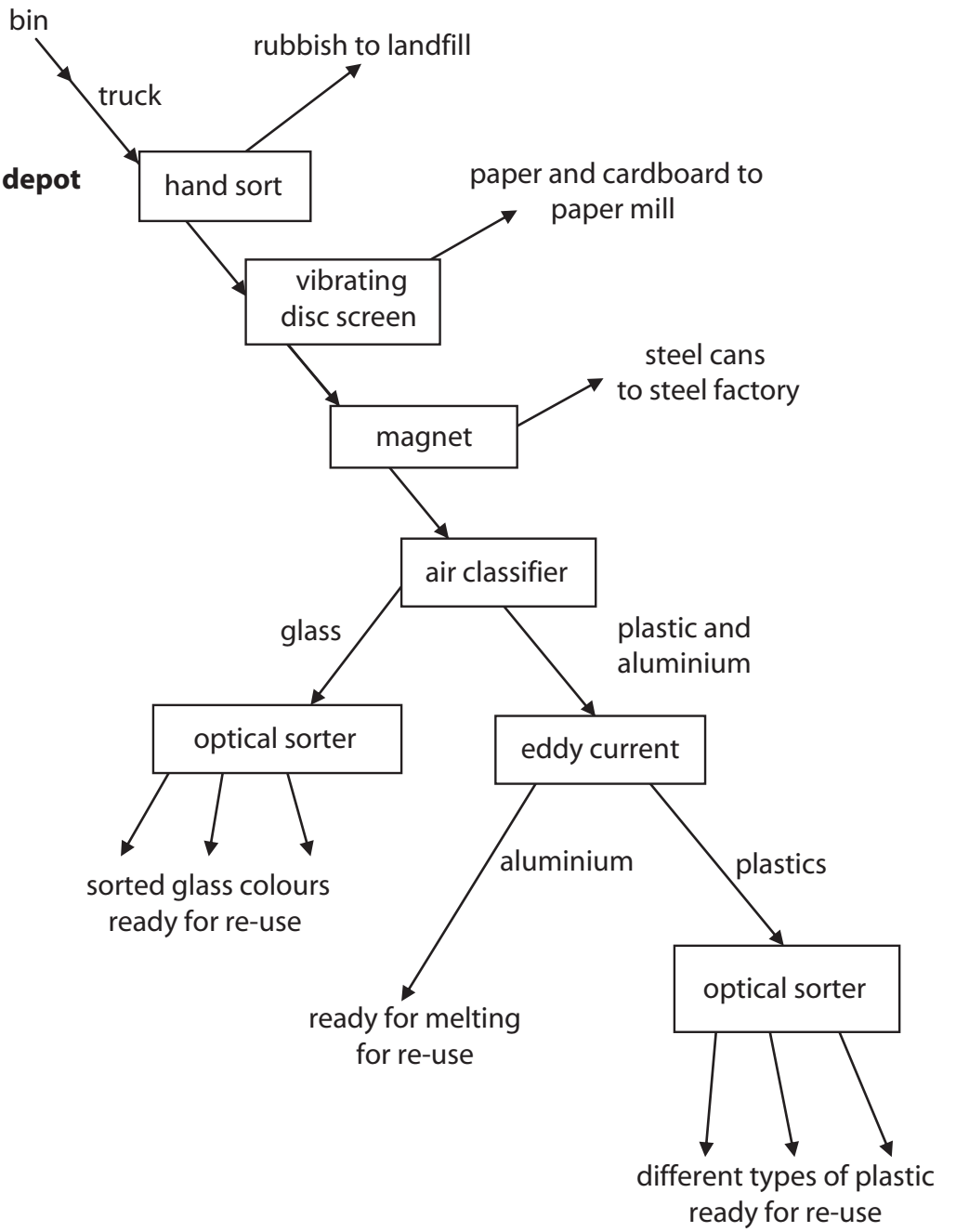
Recycling separators

After your recycling bin is collected the rubbish needs to be sorted before it can be **recycled**. Municipal councils use systems like this for sorting recycled material.



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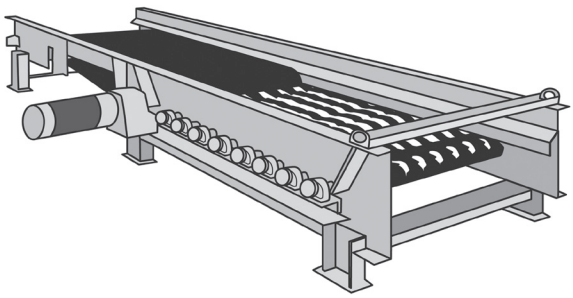
Disc screen and magnetic roller

Use the Resource sheet 'Recycling separators' to answer these questions.



☞ **1** The first step at the recycling depot is to hand sort the rubbish from the recyclables. Can you think of three items that cannot be recycled, so would need to be sent to **landfill**?

_____ _____

_____

2 After the rubbish has been removed, the conveyor takes recyclables to the vibrating disc screen where _____ and _____ are passed over the top of the discs, leaving steel and aluminium cans, bottles, jars and other plastics and glass to fall down between the discs. These containers fall onto a conveyor belt at the bottom.

3 Cardboard is **manually** separated from paper and squashed into bales. Manually means by _____.

4 List two things that can be made from recycled paper or cardboard?

_____ _____

5 As the containers leave the v_____ d_____ s_____, they pass over a **magnetic** roller that collects all of the s_____ cans while glass, aluminium and plastics continue on to the air classifier.

6 Write a paragraph that describes what you think you could do to make sorting easier and recycling more successful.

_____

_____

_____

_____

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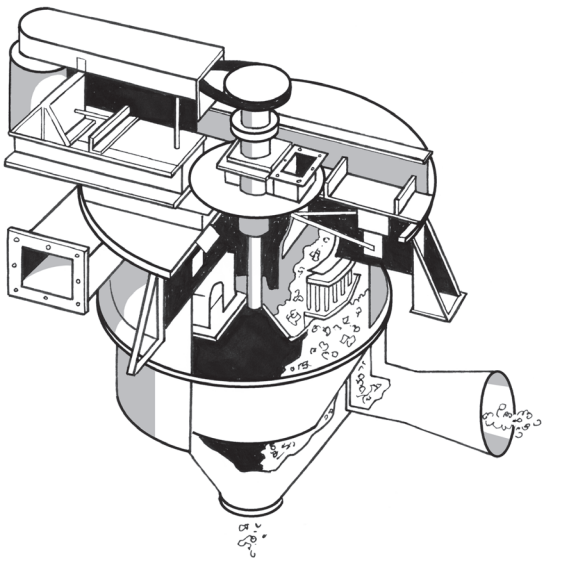
Air classifier



Look at the Resource sheet 'Recycling separators'. An air classifier separates items by creating strong blasts of air. The air lifts lighter objects such as plastic containers and aluminium cans. Glass jars and bottles are heavier, so they stay on the conveyor belt.

☞ **1** Look carefully at the diagram, then cross out the incorrect words in each sentence.

Glass containers and plastic containers and aluminium cans will leave through the top of the air classifier.

Glass containers and plastic containers and aluminium cans will leave through the bottom of the air classifier.

2 Very small pieces of broken glass are usually removed before the waste reaches the air classifier. Can you explain why this is done?

_____

_____

Density

Although we often talk about lighter and heavier objects, it is more scientific to refer to **dense** and **less dense** objects. Density measures how tightly packed an object's mass is. A pencil and a kitchen sponge weigh about the same, but a pencil is much smaller – the pencil is more dense than the sponge.

☞ **3** Cross out the incorrect item in each of the brackets.

Dense objects have their mass [tightly packed, loosely packed]. An example of a dense object is an empty [glass jar, aluminium can].

Low-density objects have their mass [tightly packed, loosely packed]. An example of a low-density object is a [glass bottle, plastic drink bottle].

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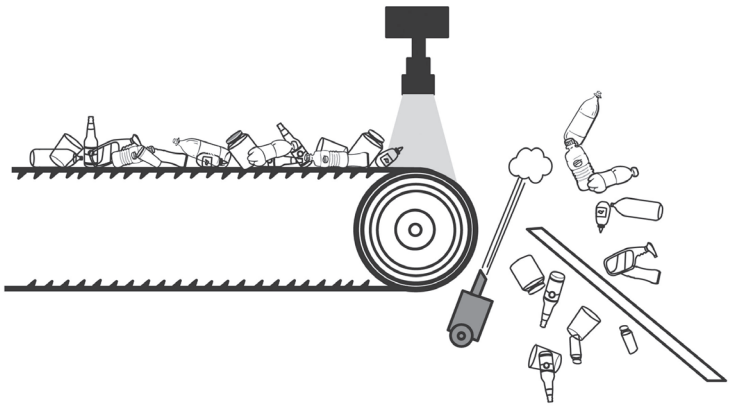
Optical sorting



After the air classifier has separated them, glass and plastic containers each go to their own **optical sorter**. The optical sorter shines a strong light on each container. A detector then decides what kind of plastic or glass it is, and as it falls, a blast of air directs it into the correct storage bin, ready to be recycled.

☞ **1** Use the word bank to complete the passage below.

Optical _____ machines use light to sort _____ and _____ into their different types and _____. Until they are sorted, they cannot be _____ into new containers. A beam of _____ shines on each piece and _____ back. Each _____ type of material reflects different colours. A _____ detector recognises each type and fires a blast of air at it to push it into the _____ storage bin.

After passing through the optical sorter, the different _____ of plastic or glass will be in _____ large bins, ready to go to the plant to be made into _____ again.

Word bank

computerised
reflects
plastics
recycled
separate
correct
light
sorting
colours
different
types
containers
glass

2 Plastic has two major advantages over glass. What are they?

_____

_____

3 Plastics that are not recyclable, such as polystyrene, have no recycling stamp on them. Have a look at a plastic bottle to see if it can be recycled. Draw the stamp that tells you that it can be.

4 What does the number in the stamp tell you? _____

_____

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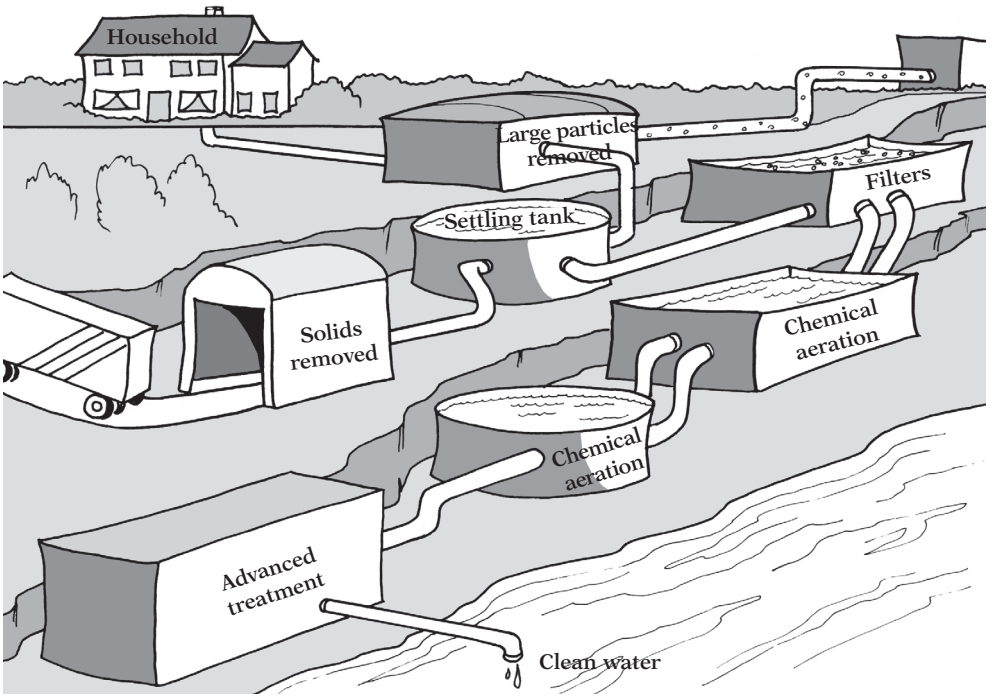
Sewerage

Waste from our homes goes into bins and down the drain to the sewer. The local council removes rubbish and garden waste. But what happens to the waste that goes into the sewer?



Sewerage refers to the system of pipes (sewers) that remove the waste and the treatment plant it goes to. **Sewage** refers to the actual waste. A lot of water is lost through sewers. Treatment plants are trying to find ways to recycle some of this water. They are also trying to reduce the amount of **nitrates** entering the sea, as these can affect some sea plants.

☞ **1** Match the sentence halves below about sewage treatment.

Bacteria are not all dangerous and	stops gases from going into the atmosphere and provides energy for power.
Water shortages have encouraged us	by not throwing rubbish items into the toilet.
Capturing the methane gas emitted by sewage	the escape of greenhouse gases, and the amount of nitrates entering the sea.
After treatment, solids can	to recycle water from sewage for watering parks and golf courses.
Water recycled from sewage is	some can be very useful to us.
I can personally reduce problems with sewage treatment	be safely used as fertilisers.
Sewerage treatment plants in Australia are finding ways to reduce water loss,	disinfected and safe to use.

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Teacher's notes – Teamwork and cooperation

Achieve!

Objectives

- To understand that cooperation between scientific and medical communities is common and necessary
- To learn that cooperation between different nations and their scientists is effective in protecting Antarctica, working on the International Space Station and controlling disease
- To know how modern disease outbreaks can be controlled by teamwork and to be aware of John Snow's work in this area in 1854

Prior knowledge

Students should be aware that scientists communicate and cooperate across national boundaries.

Background

Scientists are in constant communication, whether it be within a hospital or across a continent. Often research teams in several different countries work on the same major problem. They compete for results but they also cooperate. In particular, once a discovery has been made there is often international cooperation to make it work and improve it. The individual action taken by John Snow over cholera is no longer necessary.

Starter activity

An important factor in science is that from a local level through to an international level there is cooperation. The Activity sheet 'Antarctica' in Unit 2 gave some examples of how Antarctica is being cleaned up. Discuss with students how nations with Antarctic interests have come to an agreement to clean up the mess left from the time when discovery was considered more important than the environment. Have students brainstorm other situations where such action could be beneficial.

Activity sheets

The Activity sheet 'Reproductive technology' presents a crossword puzzle listing professions involved in IVF. This activity helps to give students an idea of scientific and other professions cooperating on a local level.

The 'Working together in Antarctica' Activity sheet looks at international scientific cooperation in Antarctica. Students order sentences describing the events involved in an imaginary accident and recovery.

'The International Space Station' looks at cooperation on the International Space Station project. Students put measurement conversion into practice to complete a passage, then sort jumbled words to complete a sentence.

The 'UNESCO' Activity sheet introduces students to this international cooperation agency. Students look at examples of UNESCO's work by selecting the correct word from each pair to complete the text.

The Activity sheet 'Cholera' introduces John Snow – known as the father of epidemiology – and explains how he stopped the spread of cholera by breaking the handle of a water pump to prevent people from using it. This example is used to illustrate the international cooperation needed for disease control in these highly mobile times. Comprehension questions are used to consolidate learning.

Extension ideas

Have students conduct a research project outlining an example of international scientific cooperation. They could research a new example or look at one of the topics covered in the unit in greater depth. The following websites provide additional information on these topics:

- www.monashivf.edu.au
- www.antarctica.gov.au
- spaceflight.nasa.gov/realdata/tracking/index.html
- www.nasa.gov/mission_pages/station/main/onthestation/facts_and_figures.html
- www.youtube.com/watch?v=r4gS9bep8Tc
- water.chemistry2011.org
- en.wikipedia.org/wiki/File:Snow-cholera-map-1.jpg

Suggested web addresses were functional at the time of publication but teachers should check them for inappropriate content and to ensure that they are still functional before use with students.

Selected answers

Reproductive technology Across: **1** pharmacologist; **3** social worker; **4** psychologist; **6** GP; **10** ethicist; **11** donor; **12** cleric; **13** lab tech; **14** midwife; **16** psychiatrist; **17** volunteer. Down: **2** obstetrician; **5** gynaecologist; **7** physiologist; **8** nurse; **9** biochemist; **15** lawyer.

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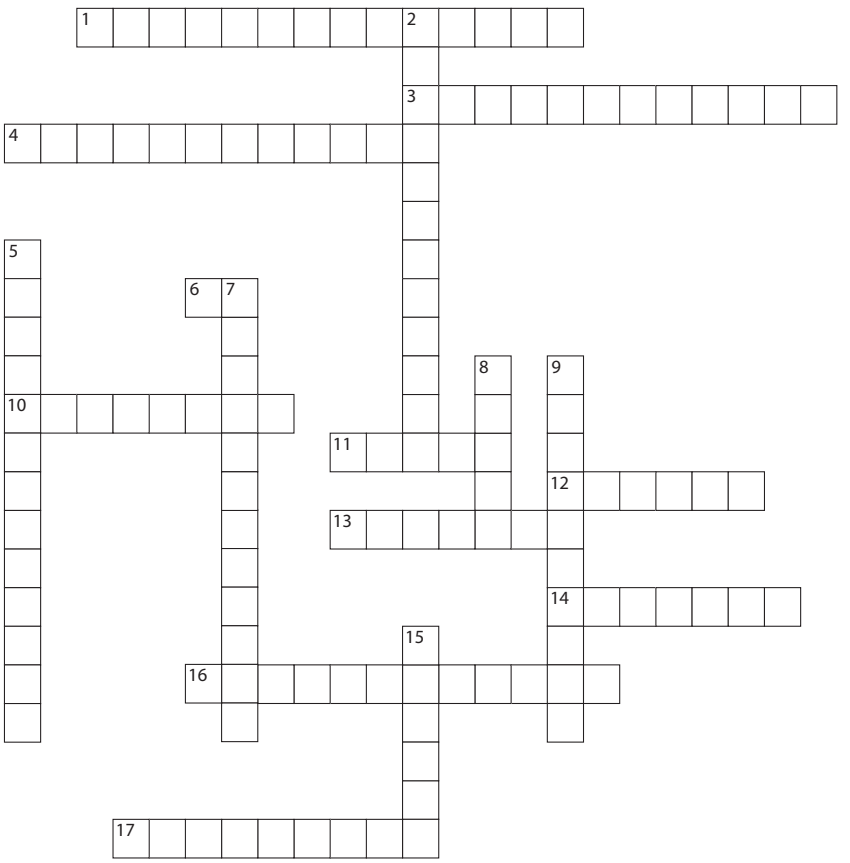
Reproductive technology

IVF – In Vitro Fertilisation – is a process used to help people have a baby. Sperm are mixed with eggs in a small glass dish. When an egg has successfully been fertilised, it is implanted into the female's uterus, where the baby develops. Many different people work together to develop and use reproductive technologies.

☞ **1** Use the word bank to complete the crossword puzzle.

Word bank

psychiatrist GP obstetrician gynaecologist nurse
pharmacologist midwife physiologist psychologist social worker
biochemist volunteer lab tech donor lawyer ethicist cleric



Across

1. Scientist who invents or improves medicines
3. Counsellor who helps people
4. Mental wellbeing and learning expert
6. Local family doctor (abbrev.)
10. Expert in discussing right or wrong
11. Person willing to give blood, sperm or eggs
12. Religious professional
13. Laboratory technician (abbrev.)
14. Nurse who manages pregnancy and birth
16. Mental health doctor
17. A person who is willing to help out with trials of treatment

Down

2. Doctor who helps with birth
5. Women's reproductive health doctor
7. Scientific expert on human body function
8. Health expert who helps patients
9. Scientist who is expert in body chemistry
15. Person who advises on legal issues

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Working together in Antarctica

Antarctica is a frozen continent that surrounds the South Pole. It is too cold for anyone to live there permanently, but it is a unique environment that fascinates many people. It has been a place of **international cooperation** for more than 60 years, with people from all over the world working together and sharing their resources and research.

☞ **1** The following fictional story parts are out of order. Number the order they should appear in for the story to make sense.

___ Four scientists were collecting ice samples on an expedition to find out more about climate change.

___ The Italians arranged for an aircraft to be put on standby at USA's McMurdo base.

___ The scientists were Linda, an Australian, Georges, a Frenchman, Luigi, an Italian, and Julio, an American.

___ He had a broken leg and his hand was crushed.

___ So, Linda contacted the Italian base by radio.

___ The scientists managed to hoist Luigi to the surface.

___ One day Dr Luigi Volta crashed into a crevasse on his snow mobile.

___ But foggy weather made that unavailable.

___ From there he was flown to Hobart, where microsurgeons repaired his hand.

___ They were working 80 km inland from Italy's Marco Zucchelli scientific base in Terra Nova Bay.

___ His leg was set and he went from Casey to France's Dumont D'Urville base by small aircraft for more treatment.

___ Instead, a helicopter picked up Luigi and flew him to Australia's Casey base for treatment.

2 Describe in your own words how scientists can help each other in Antarctica.

_____

_____

_____

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The International Space Station

Six space agencies representing 16 nations built and run the International Space Station. The ISS is an inhabited artificial satellite that orbits about 430 kilometres above the Earth's surface. Many scientific experiments are conducted on the station.

- Module length: 51 metres
- Truss length: 109 metres
- Solar array length: 73 metres
- Mass: 390 908 kilograms
- Habitable volume: 388 cubic metres
- Pressurised volume: 916 cubic metres
- Power generation: 8 solar arrays = 84 kilowatts in total
- Lines of computer code: approximately 2.3 million

Source: http://www.nasa.gov/mission_pages/station/main/onthestation/facts_and_figures.html

This information is in metric measures. Scientists all use metric measurements so that they can easily compare their information without having to convert the measurements first.

- A soccer pitch is 105 metres long by 68 metres wide.
- A singles tennis court is 24 metres long by 8 metres wide.
- A Holden Cruze diesel car has a mass of about 1500 kilograms.
- An Olympic swimming pool is about 2500 cubic metres.
- A suburban house may generate 2 kilowatts of solar power.

☞ **1** Fill in the gaps using the information above. You may wish to use a calculator. Round the answers down to the nearest whole number.

The space station module is a little over _____ tennis courts long. Its mass is equal to about _____ cars. The truss length is about _____ soccer pitches long. The pressurised volume is [more than/less than] half of an Olympic swimming pool. The space station has exactly ____ times the solar power capacity of a suburban house.

2 Sort out the jumbled nations to complete the sentence.

Most of the international space station cooperation is between the SUA (_____), PORUEE (_____), APNAJ (_____), SIARSU (_____), NAACAD (_____) and LRBAZI (_____). Eleven nations share the space agency of Europe.

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UNESCO

UNESCO stands for the United Nations Educational, Scientific and Cultural Organization. It was set up by 37 member nations in November 1945. UNESCO's mission is to promote peace and sustainable development and to reduce poverty through education, the sciences, culture, communication and information.

☞ **1** Insert the missing vowels (a, e, i, o, u) to find out about one of UNESCO's objectives.

Mobilising sc__ __nc__ kn__wl__dg__ and policy for s__st__ __n__bl__ development.

UNESCO is involved in many projects all over the world. Two are to combat global warming and water shortages, especially in poor countries. One part of a project developed especially for children is described below.

☞ **2** Choose the correct word from each pair in brackets in the passage below.

The Big Splash!

During 2011 students were invited to [avoid/participate in] what may be the biggest scientific experiment ever, by UNESCO and IUPAC (the International Union of Pure and Applied Chemistry). With their teachers, children all around the [corner/world] measured pH levels and salinity in [water/sand], filtered and purified the water and then desalinated it. Results were sent to the IUPAC website. Some disadvantaged students were also given a special opportunity to join the big experiment.

On United Nations' World Water Day on 22 March 2011, 1000 pupils from different parts of Cape Town, in South Africa, were shown the difficulties Khayelitsha slum-dwellers face daily in obtaining clean water from a standpipe. [A standpipe is like a garden tap, shared by dozens of families as their only water source.]

The children were then [transported/apparated] to Ratanga Junction to watch a play performed by the Jungle Theatre on the importance of preserving and [wasting/conserving] local water supplies.

Later, the children were handed kits to do their own [experiments/washing up] – they measured the pH of a water sample taken from Intaka Island, a wetland in Cape Town, then filtered and purified the water. For most of the children, this was the [last/first] time they had ever conducted a [culinary/chemistry] experiment. The children were very excited when they did the exercise successfully.

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Cholera

In 1854 London was suffering from an outbreak of cholera, a deadly disease. Nobody knew how the disease was spread. At that time, people had to get their water from pumps in the street. John Snow wondered if cholera was being spread by contaminated water. He made a map showing where the cases of cholera were occurring, and discovered that most cases were around Broad Street. John Snow broke the handle off the Broad Street pump so nobody could pump drinking water from it.

London's drinking water came from pools in underground beds of gravel or from underground streams. Sewage had leaked into the water, spreading cholera. John Snow is now known as the Father of Epidemiology (the study of the spread of disease). He saved dozens of lives.

☞ **1** How can natural disasters (for example, floods, cyclones, earthquakes, tsunamis) lead to cholera outbreaks?

_____

_____

2 Military Response teams and organisations such as the Red Cross often rush to disaster zones with supplies of clean drinking water. Why?

_____

In modern times when a nation reports an **epidemic**, large scale international teamwork between doctors and governments takes place to control the disease. Disease outbreaks such as SARS are controlled by international cooperation.

☞ **3** Write down four things you think governments might do to control an international disease outbreak.

_____

_____

_____

_____

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Teacher's notes – Scientific professions

Achieve!

Objectives

- To become familiar with famous Australian biological and medical scientists
- To learn that many professions and jobs require scientific knowledge and training
- To understand that scientific workers must provide justification for their decisions
- To be aware of the scientific challenges involved in developing new suburbs and houses

Prior knowledge

Students will know that some jobs are science based and require scientific qualifications. However, they may be surprised to hear that many other kinds of jobs also have scientific aspects.

Background

The scientific process is not just used by scientists. Scientific thinking and reliance on evidence is also used in many different jobs. The tools that are available to forensic scientists now mean that police and lawyers have had to increase their scientific literacy. Tasks such as planning a new suburb now include complex environmental requirements.

Starter activity

Explain to students that science is about action based on evidence. Many use it, not just professional scientists. Scientific professions such as engineering and medicine are well known, but many other jobs require scientific thinking. Hold a class brainstorm listing as many jobs as possible that might involve some form of science. Have students explain their suggestions.

Activity sheets

The Activity sheet 'Medical professions' introduces the wide variety of careers that exist in medical science. Students complete a word search introducing different vocations and some famous scientists from within those fields.

The 'Engineers' Activity sheet explains that there are many different types of engineers and they are all scientific workers. Students complete a table outlining various forms of engineering and what is involved in each field.

The Activity sheet 'Forensic scientists (1)' explains how forensic scientists gather evidence and determine what is relevant. A case study is presented and students need to determine what qualifies as useful evidence.

The 'Forensic scientists (2)' Activity sheet completes the case study outlined in 'Forensic scientists (1)' and students fill in missing vowels and complete sentences to solve the case.

The Activity sheet 'Developer of a new suburb' looks at practical applications of science, using town planning and construction as an example. Students look at a flowchart showing the planning of a new suburban development and consider where a scientist or technologist might be needed.

Extension ideas

Set a challenge – ask students to come up with jobs that do not have a need for clear scientific thinking or knowledge. When workplace safety and hygiene come into play as well as everyday job requirements, it can be difficult to find jobs without some aspect of science to them.

Have students conduct a research project on one of the scientists introduced in the word find on the 'Medical professions' Activity sheet.

For further information about forensic science, go to www.abc.net.au/science/forensic

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Medical professions

There are dozens of different careers in the field of medical science.

☞ **1** Find the words listed in the box in the word find below. Words may be written across, down, backwards and diagonally. Circle the words in the puzzle grid as you find them, and cross them off the list.

Howard Florey	Mac Burnet	Ian Frazer	John Eccles	Peter Doherty
Barry Marshall	Robin Warren	Fiona Wood	penicillin	immunity
cancer vaccine	nerves	immune defence	ulcer	helicobacter
spray-on skin	surgeon	physician	nurse	paramedic
pathologist	radiologist	physiotherapist	anaesthetist	audiologist

G D J S M E M O D K D D T B L A X L U I O T
U D L W B O N Y E R O L F D R A W O H P S D
T S I P A R E H T O I S Y H P A H D C I M O
R C I D E M A R A P O T O P V S V U G F S O
E V P K I N T H M N R X B Q Z O E O A E K W
T Z U T P N N E M E O J B W T Q L V L D T A
C L O S L O T C H K T G T E U O J C R S H N
A S H I N E B O A L K Z N I H C C K I E E O
B V I T F G D X S L S C H T B E C G R N N I
O G P E T R N R B A G B A B N Y O J I N M F
C Y N H E U E F E H X P T H Q L R C I M C Q
I T E T T S R I O S L A O V O E C L U H J E
L S E S S I R G B R W J U I C A L N K B W M
E P H E I A A N T A F F D L V I E S R U N Q
H N K A G N W H E M O A U R C D I P Q L F T
X N G N O F N Y N Y R Q E I E D U Z Y T I W
F Y R A L R I A R R M C N F P T E Q F W R H
U A L H O A B V U R N E E P H Y S I C I A N
F G M G I Z O N B A P N I K S N O Y A R P S
X S X J D E R Y C B C A Q R U X N M N X H Q
J E Q O U R Y G A E C V E Y T I N U M M I T
W V I L A F Q O M K D W Q G P T B G P G B C

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Engineers

It is not just people who work directly in science professions who use scientific methods in their work. Engineering is one example of a field that uses science. There are many different kinds of engineers. In every activity where design and construction are needed, there is a specialist engineer. A mechanical engineer designs and builds machines. A civil engineer designs and builds roads, bridges, sewers and buildings. Often, an engineer works with other specialists. A civil engineer would also work with architects, electrical engineers, and others.

☞ **1** Use the word bank to help you complete the table.

Word bank

work out ways to prevent pollution	Civil	design electronic systems
design layout of amplifiers and speakers	Aeronautical	Petroleum
car manufacture and design	Nuclear	Biomedical
design layout of coalmines	Marine	Model
find evidence at accident or crime scenes	Chemical	

Type of engineer	What they do	Type of engineer	What they do
Automotive			plan and supervise road-building
Mining			design and supervise chemical factory
	aeroplane design and maintenance	Environmental	
	design and maintain ship engines	Sound	
Electronic			design/maintain nuclear power stations
Forensic			design and build models
	find oil deposits and design oil wells		design and build artificial limbs

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Forensic scientists (1)

Forensic scientists investigate scenes of suspicious incidents, such as crimes or accidents. Forensic scientists are specialists with training.

☞ **1** Replace the missing vowels (a, e, i, o, u) in the following paragraph about what forensic scientists do.

The __v__d__nc__ they collect may be from a d__ __d body or an inj__ry. It may come from spl__shes of liquid, including bl__ __d or o__l, f__ng__rpr__nts, f__ __tpr__nts, tiny scr__ps of skin or p__ __nt, powders, b__ll__ts, sk__d m__rks, g__ns, fibres, cl__thing, make-__p – anything that is found at the sc__n__ being inv__st__g__t__d.

☞ **2** Investigating an early morning explosion in a fast food restaurant that occurred two hours after the restaurant closed, forensic scientists noticed the following things. Write whether you think each piece of evidence is useful (U) or not useful (NU). Remember, for something to be evidence, it needs to be something that shouldn't normally be there.

stack of cardboard containers		burn mark on ceiling above the safe	
uncooked meat and chicken falling out of open refrigerator		pickles and shreds of lettuce on the floor	
spatter of human blood		damage to the safe door	
single fingerprint on top of the safe – all other surfaces had been wiped		partial shoe print in the spilled lettuce and pickle	
broken window		smear of blood on the window sill	
a personnel file in the toilet			

3 After you finish Question 2, can you come up with three suspects who you think police might interview based on the evidence found – for example, someone recently sacked?

_____ _____

_____

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Forensic scientists (2)

☞ **1** Insert the missing vowels (a, e, i, o, u) to complete the passage.

After the f__st f__ __d restaurant expl__s__ __n, f__r__ns__c scientists c__ll__ct__d all of the __v__d__nc__ and returned to the l__b__r__t__ry to examine it. After two d__ys of t__sting they f__ __nd the following r__s__lts.

Collected evidence	Result of test
partial shoe print in the spilled food	made by a common running shoe, size 11, with small cut in sole
spatter of human blood and smear of blood on the window sill	from same person, not on record in national DNA database
single fingerprint on top of the safe	not in national fingerprint database
a personnel file in the toilet	file of current employee, Emma Doolan
burn mark on ceiling above the safe and damage to the safe door	simple fertiliser explosive was used – no leads to anyone

The police took statements from employees and the manager. Detective Weston summarised the statements:

- Employee Emma – not her shoe, no idea who did it, no cuts, not her fingerprint, has an alibi
- Manager – no-one has been sacked recently, matched fingerprint, last to lock up after cleaning, safe was locked, alibi checks out
- Employee Bill – no idea, no cuts, not his fingerprint, shoe size 9, has an alibi
- Employee Tracy – gossips about Emma's dumped boyfriend, Gary.

☞ **2** Use the word bank to work out what Weston decides to do next.

Officers go to check on _____. They find that he has size ____ shoes, is furious with _____ and wants to get ____ sacked from the restaurant. When the police look at his shoes there is no _____ but the police ask to see his other shoes and one has a _____ on the sole and traces of _____. They find that his blood matches the evidence and that he has a nasty _____ on his wrist. He is arrested.

Word bank

cut
evidence
Gary
gash
lettuce
her
Emma
11

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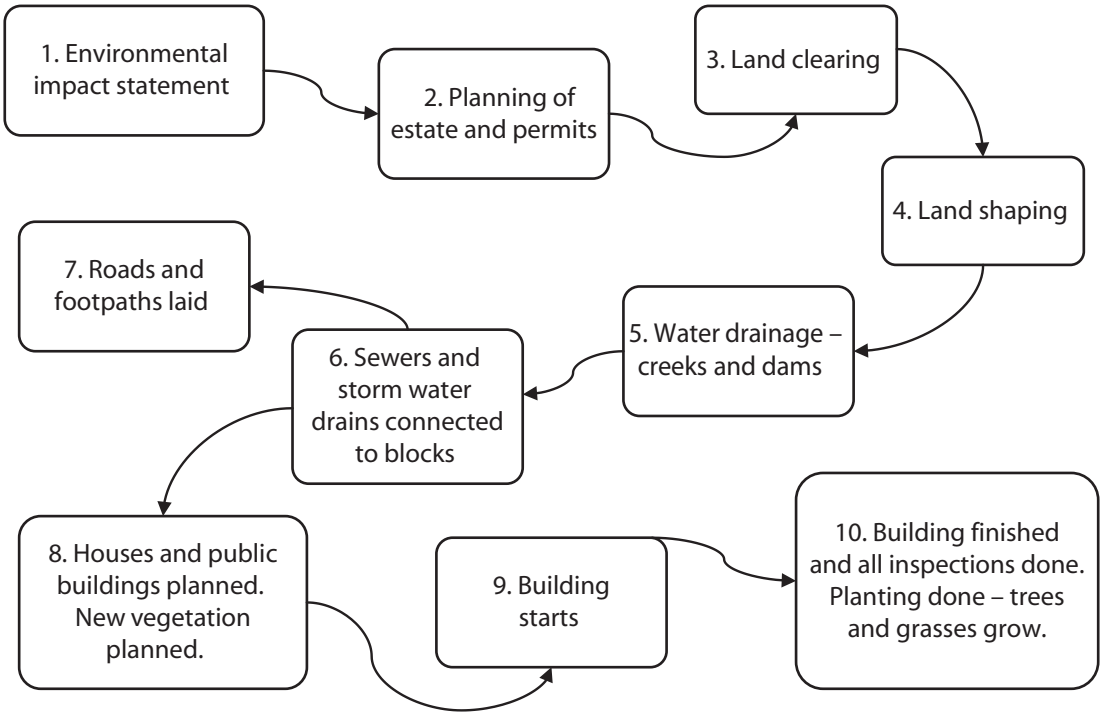
Developer of a new suburb

A developer is a person or company that clears land and organises the building of a new estate or part of a suburb. Why would a developer need scientific and technological knowledge?



☞ **1** The flowchart above shows the stages involved in planning and developing a new suburb. In the table below, list the stages that you think each person might be involved in.

Scientist or technologist	Stage number/s	Scientist or technologist	Stage number/s
civil engineer		environmental scientist	
town planner		plumbing inspector	
water engineer		horticultural scientist	
electrical inspector		sanitation engineer	

2 Fifty years ago, there were fewer rules and fewer steps involved in planning a new development. Why do you think this has changed?

_____

_____

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Teacher's notes – Farming technology

Achieve!

Objectives

- To learn that selective breeding has occurred for thousands of years
- To understand how a pedigree diagram can help trace a characteristic through generations
- To be introduced to selective breeding, artificial insemination, IVF and genetic engineering
- To learn that selective breeding has led to great increases in productivity, but has also caused some problems
- To discover how the CSIRO is trying to combat methane
- To learn the names of some unwanted feral animals and plants in Australia

Prior knowledge

Students should know the basics about reproduction. They should also have some understanding of why certain characteristics would be desirable in plants and animals.

Background

While some imported animals and plants provide much-needed crops and foods, others pose a severe danger to native animals and plants. Selective breeding has become more sophisticated over recent years, with improvements in productivity, but there are also some challenges.

Starter activity

Hold a class discussion about productivity by displaying a large and small cereal pack and a large and small milk carton. Ask students which will feed more people? Discuss how increases in population lead to greater demand for food. Ask students how we go about producing more milk, more cereal and so on. What potential problems could arise from this increased production? (Problems could include crops becoming too specialised, too much methane being produced, too much land being cleared.)

Activity sheets

The Activity sheet 'Selective breeding' provides a simple introduction to selective breeding and how techniques have changed over time. Students choose the correct word in each pair to complete the text.

The 'Breeding and pedigrees' Activity sheet presents a pedigree diagram demonstrating how breeders record particular characteristics. The text introduces the use of IVF and artificial insemination in selective breeding, and students complete sentences and match sentence halves.

The Activity sheet 'Developments in selective breeding' outlines the work of William Farrer and Sir Otto Frankel and the successes of selective breeding as well as some of the challenges. Students complete a cloze activity then fill out a table categorising various outcomes as either successes or problems. Some discussion of the value and the concerns about genetically modified foods should take place after this activity is completed.

The Activity sheet 'Greenhouse gas from farm animals' looks at methane produced by ruminants. Students complete a cloze exercise about three CSIRO projects, and select the correct word from pairs to complete a passage about a possible solution.

The 'Feral plants and animals' Activity sheet presents a word find listing the names of various feral animals and plants that exist in Australia. This activity raises awareness of the problems and the importance of quarantine in Australia. See also Activity sheet 'Quarantine laws' in Unit 10.

Extension Ideas

The breadth of activities in which Australians participate to develop and protect their country is extraordinary. Some of the rural activities will be appreciated by regional Australians and knowledge of rural life is needed by city dwellers. Any one of the above activities could lead to several open-ended research topics. More information can be found at:

- adb.anu.edu.au/biography/farrer-william-james-6145
- www.csiro.au/files/files/pq9l.pdf
- www.science.org.au/fellows/memoirs/frankel.html

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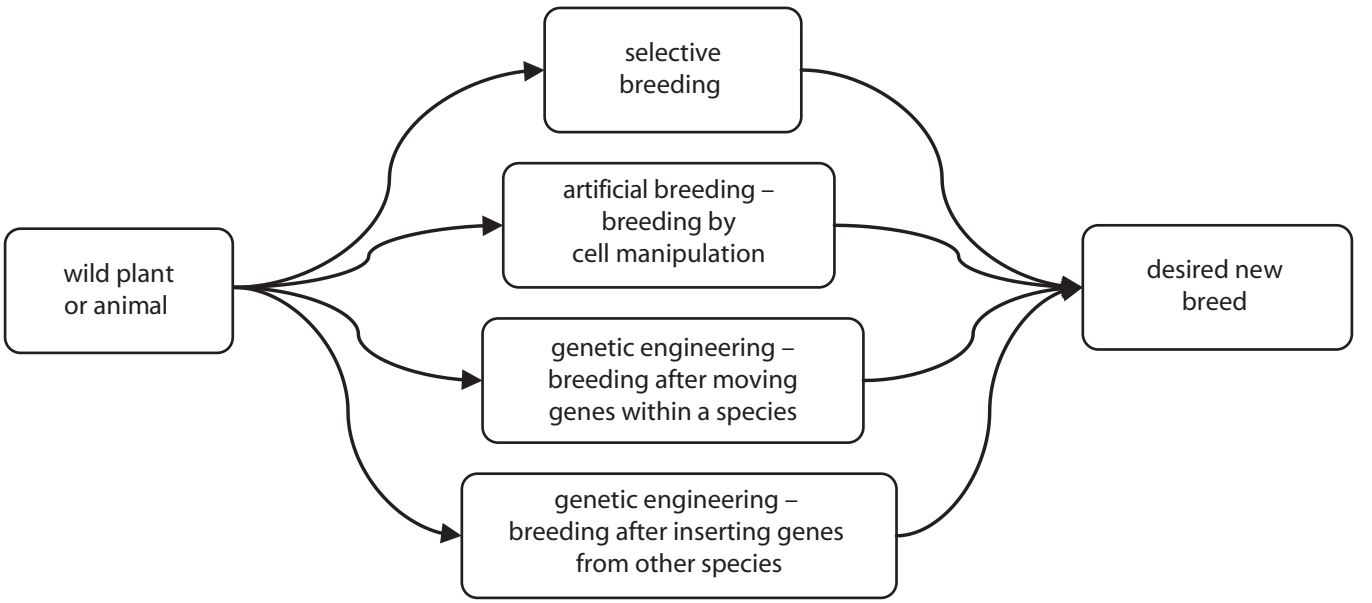
Selective breeding

Selective breeding means that a human controls which plants **reproduce** or which animals have **offspring**. The human chooses parent plants or animals with desirable features and avoids those with unwanted features. New techniques in selective breeding have been introduced over the last 60 years.



1 Circle the correct word in each pair to complete the passages below.

Selective breeding

It used to take farmers years to breed animals and plants to get [better/poorer] quality or larger **yields**. The result may have been a [smarter/fatter] sheep dog, a woollier sheep, [less/more] milk from cows, or a higher wheat yield to make more and [mouldy/better] flour.

Artificial breeding

Now [new/old] crop or animal breeds can be produced quickly [while/without] waiting for individuals to reach breeding age. Scientists can manipulate cells by fertilising [eggs/paddocks] and implanting them in the mother or [weeding/growing] plants from seeds in a laboratory.

Genetic engineering

After being removed from the mother, while the eggs of a plant or animal are [outside/beside] the mother, different nuclei or genes can be [invented/inserted]. They may come from the same [cow/species] or from different species. [Genes/Chemicals] can be added to produce certain qualities, such as insect resistance.

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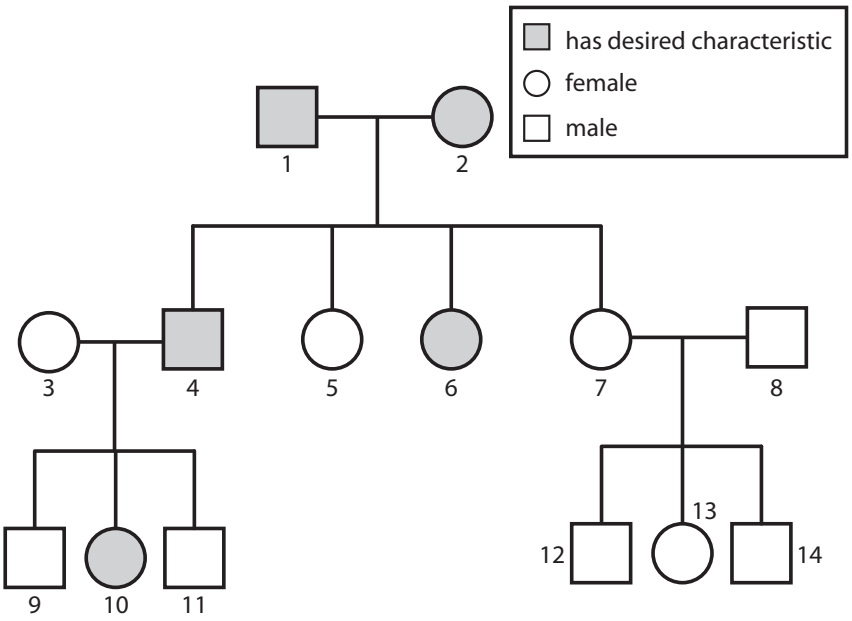
Breeding and pedigrees

Farmers want their stock to have the best **characteristics**. These characteristics are obtained through careful breeding. A **pedigree** diagram shows family relationships. It is a way of seeing how characteristics are passed on.



☞ **1** The diagram to the right shows how a particular characteristic might be present and passed on in a family line. Draw in the missing symbols in the key below.

A male is shown as	A female is shown as
A male with the characteristic is shown as	A female with the characteristic is shown as

Look at the pedigree diagram and circle the correct options in the following sentences.

2 Individual 6 is [male/female] and [shows/doesn't show] the characteristic.

3 Individual 12 is [male/female] and [shows/doesn't show] the characteristic.

4 Individual 1 is [male/female] and [shows/doesn't show] the characteristic.

5 If the characteristic is a desirable one, would you breed individual 6 with a male with the characteristic? [yes / no]

6 Draw lines to match the sentence halves.

IVF happens when an egg is fertilised by a sperm in a laboratory	and placed near the opening to the female's uterus.
Selective breeding is when a male and a female are chosen	and the fertilised egg is placed in the female's uterus.
Artificial insemination happens when sperm is taken from a male animal	for their characteristics (such as milk production) and bred together.

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Developments in selective breeding

Success with wheat breeding

William Farrer was one of the first people to improve wheat crops so they could grow in Australian conditions. Farrer developed the basis for a good wheat crop, which has heads level for harvesting, high grain yield, makes excellent flour and resists disease – especially a mould called rust.

Problem with wheat breeding

The sets of genes that plants in a crop have are almost identical. This makes them susceptible to disease. The wild originals of the plant may have genes for disease resistance that are no longer present in the crop plants.

During the mid-1900s, Russian scientist Nikolai Vavilov and others, such as CSIRO's Sir Otto Frankel, suggested that wild seeds of crops be placed in freezer storage in case we ever need to reverse the specialisation of crops by **back-breeding**.

Word bank

resistant
high
bred
rust
term
seeds
crops
disease

☞ **1** Use the word bank to complete the following sentences.

Farrer _____ wheat that was _____ yield and _____ to _____.

Crops such as grain _____ are susceptible to _____ and we guard against long-_____ problems by keeping wild _____ in cold storage.

2 The table below lists developments that have occurred as a result of selective breeding. Write whether you think each development is a success or a problem.

Development	Success or problem?
Much higher yielding rice has been developed.	
Rice that grows with less water has been bred.	
Death at birth in some farm animals is increasing.	
Scientists can produce genetically modified food crops.	

3 Why might selective breeding lead to problems at birth? _____

_____

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Greenhouse gas from farm animals

Animals that chew cud, such as cattle and sheep, are called ruminants. A part of their intestines called the rumen, which is one of four stomachs, contains billions of bacteria and protozoa that help to break up grass into useful nutrients. This produces **methane gas**, which is a **greenhouse gas**, as a by-product. Greenhouse gas from farm animals accounts for about 10 per cent of Australia's greenhouse gas emissions. CSIRO scientists are trying to find ways to reduce emissions of methane gas from cattle and sheep. They have three projects underway:

- study of **gut microbes** in ruminants
- measuring methane emissions from cattle
- investigating how anti-microbial properties contained in different foods affect the microbes in the rumen.

☞ **1** Fill in the missing words to complete the sentences below.

Sheep and cattle produce _____ – ____ % of all of the greenhouse gas emissions in Australia. Some CSIRO scientists are studying g___ m_____ in ruminants. Other scientists are measuring _____ emissions from cattle in Northern Australia, while some investigate _____ properties of different foods.

Tammar Wallaby

2 Choose the correct word from each pair to complete the sentences below.

Native [grazers/predators] such as the Tammar Wallaby have **microbes** in their [pouch/foregut] that break down grasses differently and produce far less methane than ruminants. Scientists are [hopeful/frightened] that this knowledge will help them to [increase/reduce] methane production in cattle and sheep in future.

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Feral plants and animals

When plant and animal species from other places are introduced into Australia they can threaten our unique wildlife, which has little defence against introduced species. Feral imported species such as rabbits and foxes can cause huge damage.

Australia is isolated – only Indonesia and Papua New Guinea are nearby. This means that few non-native animals and plants have found their way to Australia by themselves. However, some have been brought in by humans, either intentionally or accidentally. The Australian quarantine service [AQIS] works to prevent any new species from being introduced into Australia and harming the environment.

☞ **1** The following animals and plants have been introduced to Australia and become a problem. Find their names in the word find below. Words may be written across, down, backwards and diagonally. Circle the words in the puzzle grid as you find them, and cross them off the list.

Salvation Jane	rabbit	mosquitofish	blackbird	camel	bridal creeper
Paterson's curse	cane toad	goldfish	cat	goat	agapanthus
crown-of-thorns	carp	starling	dog	lantana	fox

P	V	U	A	Y	C	S	F	V	U	P	G	D	I	F	C	M	Y	E	S
A	O	F	O	F	A	Y	K	B	B	N	C	D	I	J	S	Y	W	S	E
T	A	R	M	M	O	S	Q	U	I	T	O	F	I	S	H	N	H	H	F
E	P	F	G	W	G	T	Q	L	U	X	T	G	F	H	E	I	Y	P	G
R	V	L	O	T	D	L	R	P	D	O	G	Q	Q	I	D	T	U	U	F
S	F	U	L	G	E	A	C	R	O	W	N	O	F	T	H	O	R	N	S
O	Y	Y	D	E	T	A	S	A	L	V	A	T	I	O	N	J	A	N	E
N	U	B	F	S	B	A	B	W	R	A	B	B	I	T	C	A	T	L	F
S	V	A	I	X	Y	T	R	J	O	S	D	L	D	U	J	S	T	X	I
C	A	U	S	Q	U	X	I	K	H	O	T	A	Q	W	H	A	J	E	H
U	S	C	H	W	E	E	D	O	L	Z	Q	C	E	H	E	G	B	U	G
R	O	T	S	G	U	P	A	Q	U	V	B	K	C	A	G	A	C	T	K
S	E	L	Q	U	I	V	L	P	O	W	D	B	E	A	H	P	C	O	V
E	M	C	A	M	E	L	C	H	G	A	X	I	F	G	C	A	G	N	L
G	M	B	A	C	R	U	R	K	O	B	L	R	Z	C	O	N	V	S	A
X	A	K	N	K	A	A	E	T	A	I	I	D	X	C	T	T	K	O	N
Z	L	D	S	H	S	R	E	Q	T	F	Z	E	W	J	J	H	T	Y	T
M	D	J	J	E	A	N	P	M	O	N	W	S	X	X	J	U	R	V	A
N	J	A	D	T	A	G	E	N	W	B	M	J	O	J	A	S	B	W	N
E	W	I	E	C	B	A	R	S	Q	N	S	F	X	A	U	Y	S	I	A

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Teacher's notes – Addressing environmental concerns

Achieve!

Objectives

- To learn how environmental protection requires scientific action and thought
- To understand the importance of quarantine laws
- To learn to view media reports about the environment critically
- To understand that environmental problems can be linked and that addressing one can worsen another
- To know how individual daily actions can contribute to environmental problems

Prior knowledge

Students should be aware that there are personal and local environmental issues as well as regional and global issues.

Background

People in all walks of life can operate scientifically and act responsibly towards the environment. Environmental issues mean life and death to some people in the world. It is important to get all the facts about environmental issues, and for each individual to examine their own behaviour to avoid harmful activity.

Starter activity

Display the table on the Activity sheet 'Who works scientifically?' and have a class discussion about behaviours that people thinking scientifically might demonstrate. Ask students which of these qualities they recognise in themselves.

Activity sheets

The Activity sheet 'Who works scientifically?' introduces the behaviours associated with scientific thinking. Students consider the scientific approaches involved in a variety of careers.

The 'Quarantine laws' Activity sheet explains that Australian flora and fauna are unique and outlines the challenges involved in protecting them. Students answer a series of true/false questions exploring the issue. See also Activity sheet 'Feral plants and animals' from Unit 9.

The Activity sheet 'Climate change in the news' examines reporting of information in the media. Students examine three mock news articles about a local event and a global event with local connotations in order to develop critical reading skills. See also the Activity sheet 'Immunisation and the media' in Unit 5 for a view of reporting on vaccines.

'Doing the right thing?' looks at how, with the best of intentions, we can still harm the environment. Students complete a table to demonstrate their understanding of the implications of well-intended actions.

The Activity sheet 'Student waste' encourages students to identify wasteful activities in a diary entry about a day at school.

Extension ideas

- Students could keep an envirodiary for a day or week, noting any actions that either harm or help the environment.
- Students could assess their desired vocation with respect to scientific thought and action.

Selected answers

Doing the right thing **1** wastes drinking water, **2** use clean, renewable energy to power the plant **3** droppings can get washed into waterways, causing pollution **4** fire danger **5** should have chosen a day with no fire restrictions **6** smoke/pollution **7** no bagged goods in recycling **8** put bottles into recycling bin loose; put bags in rubbish or re-use them as bin liners.

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Who works scientifically?

It is not only scientists who think and work scientifically. The character traits and behaviour of scientists are qualities and skills that are needed in all different kinds of work.

Scientists need to:

1. be curious
2. be observant
3. keep records
4. rely on evidence rather than hearsay or old beliefs
5. measure accurately
6. check hypotheses by testing
7. think of explanations and test them
8. improve explanations when new evidence is discovered
9. collect and share evidence
10. encourage others to check for accuracy
11. communicate
12. have passion for their work
13. be patient and accurate
14. be creative in seeking answers
15. care about the effects of their work

☞ **1** Look at the list of careers below. List the numbers of the qualities listed above that you think would help in each career.

2 Tick any careers that have more than three qualities listed.

- ☐ Motor mechanic _____
- ☐ Fast food restaurant manager _____
- ☐ Librarian _____
- ☐ Factory safety officer _____
- ☐ Police officer _____
- ☐ Accountant _____
- ☐ Child care worker _____
- ☐ Teacher _____
- ☐ Building site worker _____
- ☐ Nurse _____

3 Which of the careers received a tick?

_____

_____

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Quarantine laws

Australia has very strict laws about what you can bring into the country. These laws are to protect our native plants and animals and to keep our crops and farm animals disease-free. Every entry point into Australia, for both people and cargo, has Australian Quarantine and Inspection Service (AQIS) inspectors and sniffer dogs. Their job is to make sure that no foreign species are introduced into Australia.

☞ **1** Circle 'true' or 'false' for each of the following statements.

Many Australian native animals and plants have little defence against infections.	True	False
Some living things have crossed seas and oceans and reached other countries in the past.	True	False
All of our native animals could survive attack by predators.	True	False
Animals that have been imported in the past, such as cats and foxes, have done no harm.	True	False
Every Australian citizen should respect quarantine laws.	True	False
There are no diseases found in crops and farm animals that we do not already have in Australia.	True	False
On other continents, almost all marsupials and all monotremes have died out.	True	False
A wooden toy or a salami could never cause serious problems by bringing in insect eggs or worms.	True	False
People who try to avoid quarantine are criminals who may cause death to native plants or animals, or to an important crop or herd.	True	False

2 Write a paragraph describing your opinion about one of the statements above.

_____

_____

_____

_____

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Climate change in the news

Consider the following mock news articles.

Article 1: Local council builds higher sea wall

Rocky Cove's sea wall will increase in height from 1 metre to 12 metres. Council spokesperson Irma Pearce said the council takes climate change seriously. Local residents are concerned that the wall will block their views. Hotelier Iva Grigg said his rooms and dining terrace would look out to a concrete wall. A science teacher at Cove College said the wall was an overreaction.

Article 2: Sceptics rally

Rocky Cove residents who do not believe in climate change held a rally at the sea wall. Hilda Thorne said that some members had actually seen the water level dropping twice a day and that there had been very cool weather recently. A new wall was not needed.

Article 3: It's not all FUN

Tuvalu residents are worried about their future. A press conference of leaders today at Funafuti Airport (destination code FUN) said that the low-lying coral islands may become permanently flooded if water levels rise by a metre over the next 100 years. Global warming may just be a distant concern to the rest of the world, but in Tuvalu it is looming as a disaster.

☞ **1** Insert the missing vowels (a, e, i, o, u), then match the sentence halves.

The s__ __ wall extension is	and there will still be h__t and cold days.
The dr__pp__ng water l__v__l was	given a more detailed __nsw__r.
Cl__m__t__ change is a long t__rm thing	n__c__ss__ry change.
Tuvalu pe__pl__ are right to be w__rr__ __d	not n__ __d__d yet, and m__ch too high.
S__m__ people will be ups__t about	about a 1 m__tr__ change in 100 y__ __rs.
The sc__ence teacher sh__ __ld have	h__lp either s__d__ of a dispute.
Over-r__ __ct__ __ns do not	d__e to tides and not climate ch__ng__.

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Doing the right thing?

Sometimes people try to help the environment, but end up causing more problems.

☞ **1** Fill in the gaps in the table to explain where these people went wrong, and what they should have done instead.

Action	Problem[s]	Should have
Rinsed food cans for recycling under running water.	**1**	rinsed them in used dishwater or grey water from clothes washing.
Used millions of tonnes of coal to generate electricity to power a desalination plant.	Burning coal releases a huge amount of carbon dioxide into the atmosphere.	**2**
Swept dog droppings off the driveway into the gutter.	**3**	picked the droppings up and put them in the bin.
Showed tourists from overseas the Australian bush and lit a portable BBQ on a total fire ban day.	**4**	**5**
Pruned large garden trees and burned the branches on a bonfire.	**6**	delivered the branches to the council for mulching.
Neatly re-used old plastic bags to wrap bottles for the recycling bin.	**7**	**8**

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Student waste

There are lots of different ways we can be wasteful.

a) Littering
b) Overuse of packaging
c) Waste of chemicals, e.g. soap
d) Waste of water
e) Throwing recyclable items in the rubbish instead of recycling
f) Waste of fuel
g) Waste of paper
h) Waste of reusable containers

☞ **1** Read the diary entry below. Next to each action, write the letters of the different kinds of waste that apply.

Today I climbed out of bed and put my pyjamas in the wash. I'd only worn them once, but I like wearing clean PJs every night. _____ I had cereal from a single-serve box (love the variety pack!) and threw the box in the rubbish bin. _____ I took a 20-minute shower – hey, I was dirty – _____ then mum drove me the 500 metres to school. _____ I had some chips then dropped the packet on the ground. _____ I struggled in Art today – it just wasn't happening for me. I ended up chucking 12 bits of paper in the bin. _____ I left my lunch wrap on the ground, but I threw the plastic container in the rubbish. _____ After school mum picked me up. _____

2 Now rewrite the diary entry with no wasteful actions.

_____

_____

_____

_____

_____

_____

_____

_____

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ABN 50 074 266 023
108 Main Rd Clayton South VIC 3169
+61 3 9558 4433
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